With Cage

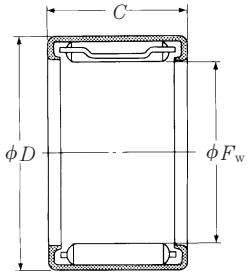
FJ • FJH (Open Type)

FJL

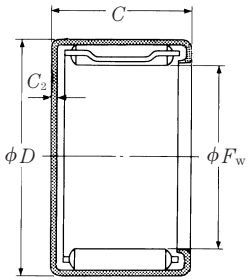
MFJ • MFJH (Closed Type)

MFJL

Inscribed Circle Diameter 10 — 16 mm



Open Type



Closed Type

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min^{-1})
F_w	D	C $^{0}_{-0.25}$	C_2 max	C_r		P_{max}		Oil
10	14	10	0.8	3 500	360	1 760	179	32 000
	14	15	—	5 550	570	3 150	325	32 000
	16	10	1.0	4 900	500	2 100	214	34 000
	17	10	1.0	4 900	500	1 990	203	36 000
	17	12	1.0	5 700	580	2 410	246	36 000
	17	15	1.0	7 850	800	3 650	375	36 000
	17	20	1.0	11 200	1 140	5 750	585	36 000
12	16	10	0.8	4 150	420	2 210	225	26 000
	18	12	1.0	6 450	655	3 050	310	28 000
	19	12	1.0	6 650	675	2 950	300	28 000
	19	15	1.0	9 200	935	4 500	455	28 000
	19	20	1.0	13 000	1 330	7 000	715	28 000
	19	25	1.0	16 600	1 690	9 550	975	28 000
13	19	12	1.0	6 950	710	3 400	345	26 000
14	20	12	1.0	6 500	665	3 250	335	24 000
	20	16	1.0	9 500	970	5 300	540	24 000
	22	16	1.1	11 600	1 180	5 700	580	24 000
	22	20	1.1	15 100	1 540	8 050	820	24 000
15	21	10	—	5 850	595	2 770	283	22 000
	21	12	1.0	7 650	780	3 900	400	22 000
	21	16	1.0	11 000	1 120	6 200	635	22 000
	22	10	1.0	6 450	655	2 880	294	22 000
	22	12	1.0	7 450	760	3 500	355	22 000
	22	15	1.0	10 300	1 050	5 300	540	22 000
	22	20	1.0	14 700	1 500	8 300	845	22 000
	22	25	1.0	18 700	1 900	11 300	1 150	22 000
16	22	12	1.0	7 100	725	3 750	380	20 000
	22	16	1.0	10 400	1 060	6 050	620	20 000
	24	16	1.1	13 100	1 330	6 700	685	20 000
	24	20	1.1	17 100	1 740	9 500	970	20 000

Bearing Numbers				Mass (g)		Matching Inner Rings	
Open		Closed		approx			
FJ FJH	FJL	MFJ MFJH	MFJL	Open	Closed		
FJ-1010		**MFJ-1010**		3.6	4.1	**FIR-071010**	—
FJ-1015L		—		5.4	—	**FIR-071015**	—
FJH-1010		**MFJH-1010**		6.1	6.6	**FIR-071010**	—
	FJL-1010		**MFJL-1010**	7.0	7.8	**FIR-071010**	—
	FJL-1012L		**MFJL-1012L**	8.4	9.3	**FIR-071012**	—
	FJL-1015		**MFJL-1015**	11	12	**FIR-071015**	—
	FJL-1020		**MFJL-1020**	14	16	—	—
FJ-1210		**MFJ-1210**		4.1	4.5	**FIR-081210**	—
FJH-1212		**MFJH-1212**		7.7	9.0	**FIR-081212**	—
	FJL-1212L		**MFJL-1212L**	9.8	11	**FIR-081212**	—
	FJL-1215		**MFJL-1215**	12	14	**FIR-081215**	—
	FJL-1220		**MFJL-1220**	16	19	—	—
	FJL-1225L		**MFJL-1225L**	20	24	—	—
FJ-1312		**MFJ-1312**		8.6	9.5	**FIR-101312**	—
FJ-1412		**MFJ-1412**		10	11	**FIR-101412**	—
FJ-1416		**MFJ-1416**		13	14	**FIR-101416**	—
	FJL-1416		**MFJL-1416L**	17	19	**FIR-101416**	—
	FJL-1420L		**MFJL-1420L**	21	24	**FIR-101420**	—
FJ-1510		—		8.1	—	**FIR-101510**	—
FJ-1512		**MFJ-1512**		10	11	**FIR-101512**	**FIR-121512**
FJ-1516		**MFJ-1516**		13	14	**FIR-121516**	—
	FJL-1510L		**MFJL-1510L**	9.8	11	**FIR-101510**	—
	FJL-1512L		**MFJL-1512L**	12	13	**FIR-101512**	**FIR-121512**
	FJL-1515		**MFJL-1515**	14	16	**FIR-101515**	—
	FJL-1520		**MFJL-1520**	20	22	**FIR-101520**	—
	FJL-1525L		**MFJL-1525L**	24	27	**FIR-101525**	—
FJ-1612		**MFJ-1612**		11	12	**FIR-121612**	—
FJ-1616		**MFJ-1616**		14	15	**FIR-121616**	—
	FJL-1616L		**MFJL-1616L**	19	21	**FIR-121616**	—
	FJL-1620L		**MFJL-1620L**	24	27	**FIR-121620**	—

With Cage

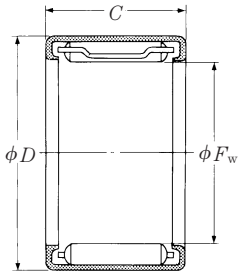
FJ (Open Type)

FJL

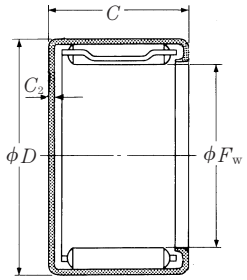
MFJ (Closed Type)

MFJL

Inscribed Circle Diameter 17 — 25 mm



Open Type



Closed Type

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min⁻¹)
F_w	D	C $^{0}_{-0.25}$	C_2 max	C_r		P_{max}		Oil
17	23	12	1.0	8 450	860	4 450	455	19 000
	23	16	1.0	12 100	1 230	7 100	720	19 000
	24	15	1.0	10 800	1 110	5 700	580	20 000
	24	20	1.0	15 400	1 570	8 950	910	20 000
	24	25	1.0	19 600	2 000	12 200	1 240	20 000
18	24	12	1.0	7 650	780	4 200	430	18 000
	24	16	1.0	11 200	1 140	6 800	695	18 000
	25	13	1.0	9 300	945	4 700	480	18 000
	25	15	1.0	11 400	1 160	6 100	620	18 000
	25	17	1.0	13 300	1 360	7 500	765	18 000
	25	20	1.0	16 100	1 640	9 600	975	18 000
	25	25	1.0	20 500	2 090	13 100	1 330	18 000
20	26	12	1.0	8 150	835	4 650	475	16 000
	26	16	1.0	11 900	1 210	7 550	770	16 000
	26	20	1.0	15 300	1 560	10 500	1 070	16 000
	27	15	1.0	12 300	1 260	6 900	705	16 000
	27	20	1.0	17 500	1 790	10 900	1 110	16 000
	27	25	1.0	22 300	2 270	14 800	1 510	16 000
	27	30	1.0	26 800	2 730	18 700	1 910	16 000
22	28	12	1.0	8 650	880	5 150	525	14 000
	28	16	1.0	12 600	1 290	8 350	850	14 000
	28	20	1.0	16 200	1 660	11 500	1 180	14 000
	29	10	1.0	7 950	810	4 000	405	15 000
	29	15	1.0	12 700	1 300	7 300	745	15 000
	29	20	1.0	18 100	1 840	11 500	1 170	15 000
	29	25	1.0	23 000	2 350	15 700	1 600	15 000
	29	30	1.0	27 700	2 820	19 800	2 020	15 000
25	32	16	1.0	15 200	1 550	9 350	955	13 000
	32	20	1.0	19 800	2 020	13 100	1 340	13 000
	32	26	1.0	26 200	2 670	18 800	1 920	13 000
	33	10	1.0	8 800	895	4 300	435	13 000
	33	15	1.0	15 400	1 570	8 800	895	13 000
	33	20	1.0	21 900	2 240	13 900	1 420	13 000
	33	25	1.0	28 000	2 850	19 000	1 940	13 000
	33	30	1.0	33 500	3 450	24 100	2 460	13 000

Bearing Numbers				Mass (g)		Matching Inner Rings	
Open		Closed		approx			
FJ	FJL	MFJ	MFJL	Open	Closed		
FJ-1712		**MFJ-1712**		10	11	—	—
FJ-1716		**MFJ-1716**		14	16	—	—
	FJL-1715		**MFJL-1715**	16	18	**FIR-121715**	—
	FJL-1720L		**MFJL-1720L**	21	24	**FIR-121720**	—
	FJL-1725L		**MFJL-1725L**	27	30	**FIR-121725**	—
FJ-1812		**MFJ-1812**		12	14	**FIR-151812**	—
FJ-1816		**MFJ-1816**		16	18	**FIR-151816**	—
	FJL-1813		**MFJL-1813**	14	16	—	—
	FJL-1815		**MFJL-1815**	17	19	**FIR-151815**	—
	FJL-1817L		**MFJL-1817L**	20	22	—	—
	FJL-1820L		**MFJL-1820L**	22	25	**FIR-151820**	—
	FJL-1825		**MFJL-1825**	28	31	**FIR-151825**	—
FJ-2012		**MFJ-2012**		13	15	**FIR-172012**	—
FJ-2016		**MFJ-2016**		17	19	**FIR-172016**	—
FJ-2020		**MFJ-2020**		22	24	**FIR-152020**	**FIR-172020**
	FJL-2015L		**MFJL-2015**	19	21	**FIR-152015**	—
	FJL-2020		**MFJL-2020**	24	27	**FIR-152020**	**FIR-172020**
	FJL-2025L		**MFJL-2025L**	30	33	**FIR-152025**	—
	FJL-2030L		**MFJL-2030L**	37	41	**FIR-152030**	—
FJ-2212		**MFJ-2212**		14	17	**FIR-172212**	—
FJ-2216		**MFJ-2216**		19	22	**FIR-172216**	—
FJ-2220		**MFJ-2220**		23	26	**FIR-172220**	—
	FJL-2210L		**MFJL-2210L**	13	15	**FIR-172210**	—
	FJL-2215L		**MFJL-2215L**	20	22	**FIR-172215**	—
	FJL-2220L		**MFJL-2220L**	27	30	**FIR-172220**	—
	FJL-2225L		**MFJL-2225L**	33	37	**FIR-172225**	—
	FJL-2230L		**MFJL-2230L**	40	44	**FIR-172230**	—
FJ-2516		**MFJ-2516**		24	27	**FIR-202516**	—
FJ-2520		**MFJ-2520**		31	34	**FIR-202520**	—
FJ-2526		**MFJ-2526**		40	43	**FIR-202526**	—
	FJL-2510		**MFJL-2510**	17	19	**FIR-202510**	—
	FJL-2515		**MFJL-2515L**	26	29	**FIR-202515**	—
	FJL-2520		**MFJL-2520**	35	39	**FIR-202520**	—
	FJL-2525L		**MFJL-2525L**	43	48	**FIR-202525**	—
	FJL-2530		**MFJL-2530**	52	58	**FIR-202530**	—

With Cage

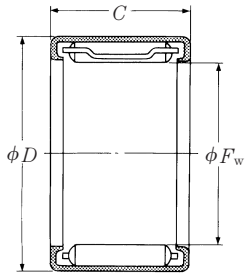
FJ (Open Type)

FJL

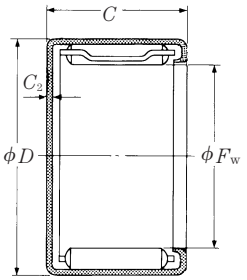
MFJ (Closed Type)

MFJL

Inscribed Circle Diameter 28 — 40 mm



Open Type



Closed Type

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min^{-1})
F_w	D	C $^{0}_{-0.25}$	C_2 max	C_r		P_{max}		Oil
28	35	16	1.0	15 600	1 590	9 950	1 020	11 000
	35	20	1.0	20 500	2 090	14 200	1 450	11 000
	35	26	1.0	26 900	2 750	20 200	2 060	11 000
	37	20	1.0	25 400	2 590	15 500	1 580	11 000
	37	30	1.0	39 000	4 000	27 100	2 760	11 000
30	37	16	1.0	15 600	1 590	10 100	1 030	10 000
	37	20	1.0	19 400	1 970	13 300	1 360	10 000
	37	26	1.0	26 000	2 660	19 500	1 990	10 000
	40	15	1.3	17 200	1 750	9 100	925	11 000
	40	20	1.3	25 800	2 630	15 300	1 560	11 000
	40	25	1.3	33 500	3 450	21 600	2 200	11 000
	40	30	1.3	41 000	4 200	27 800	2 840	11 000
32	42	20	1.3	26 700	2 720	16 200	1 650	10 000
	42	30	1.3	42 500	4 300	29 400	3 000	10 000
35	42	12	1.0	12 000	1 230	7 550	770	9 000
	42	16	1.0	18 100	1 850	12 800	1 300	9 000
	42	20	1.0	23 600	2 410	17 900	1 830	9 000
	42	26	1.0	31 500	3 200	25 800	2 630	9 000
	45	12	1.3	13 200	1 350	6 650	680	9 000
	45	15	1.3	18 200	1 860	10 100	1 030	9 000
	45	20	1.3	27 400	2 790	17 000	1 740	9 000
	45	25	1.3	36 000	3 650	24 000	2 450	9 000
	45	30	1.3	43 500	4 450	31 000	3 150	9 000
40	47	16	1.0	18 600	1 890	13 600	1 390	7 500
	47	20	1.0	23 500	2 400	18 500	1 890	7 500
	47	26	1.0	31 500	3 200	26 900	2 740	7 500
	50	15	1.3	19 900	2 030	11 600	1 180	8 000
	50	20	1.3	29 900	3 050	19 600	2 000	8 000
	50	25	1.3	39 000	3 950	27 600	2 810	8 000
	50	30	1.3	47 500	4 850	35 500	3 650	8 000

Bearing Numbers				Mass (g)		Matching Inner Rings	
Open		Closed		approx			
FJ	FJL	MFJ	MFJL	Open	Closed		
FJ-2816		**MFJ-2816**		27	31	**FIR-222816**	—
FJ-2820		**MFJ-2820**		34	38	**FIR-222820**	—
FJ-2826		**MFJ-2826**		45	49	**FIR-222826**	—
	FJL-2820L		**MFJL-2820L**	44	48	**FIR-222820**	—
	FJL-2830L		**MFJL-2830L**	65	72	**FIR-222830**	—
FJ-3016L		**MFJ-3016L**		26	31	—	—
FJ-3020		**MFJ-3020**		35	39	**FIR-253020**	—
FJ-3026		**MFJ-3026**		46	50	**FIR-253026**	—
	FJL-3015L		**MFJL-3015L**	49	55	**FIR-253015**	—
	FJL-3020L		**MFJL-3020L**	52	58	**FIR-253020**	—
	FJL-3025L		**MFJL-3025L**	65	72	**FIR-253025**	—
	FJL-3030L		**MFJL-3030L**	78	86	**FIR-253030**	—
	FJL-3220		**MFJL-3220**	55	62	**FIR-283220**	—
	FJL-3230		**MFJL-3230**	82	91	**FIR-283230**	—
FJ-3512		**MFJ-3512**		24.3	30	**FIR-303512**	—
FJ-3516		**MFJ-3516**		32	38	—	—
FJ-3520		**MFJ-3520**		41	45	**FIR-303520**	—
FJ-3526		**MFJ-3526**		54	58	**FIR-303526**	—
	FJL-3512		**MFJL-3512**	36	40	**FIR-303512**	—
	FJL-3515		**MFJL-3515**	44	49	**FIR-303515**	—
	FJL-3520L		**MFJL-3520L**	60	67	**FIR-303520**	—
	FJL-3525L		**MFJL-3525L**	74	82	**FIR-303525**	—
	FJL-3530L		**MFJL-3530L**	90	100	**FIR-303530**	—
FJ-4016		**MFJ-4016**		34	43	—	—
FJ-4020		**MFJ-4020**		46	51	**FIR-354020**	—
FJ-4026		**MFJ-4026**		60	65	**FIR-354026**	—
	FJL-4015L		**MFJL-4015L**	50	55	**FIR-354015**	—
	FJL-4020L		**MFJL-4020L**	67	74	**FIR-354020**	—
	FJL-4025L		**MFJL-4025L**	84	93	**FIR-354025**	—
	FJL-4030L		**MFJL-4030L**	100	110	**FIR-354030**	—

With Cage

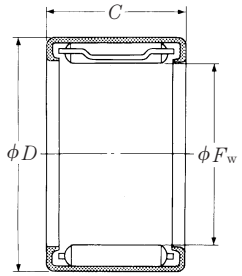
FJ (Open Type)

FJL

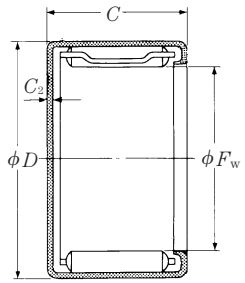
MFJ (Closed Type)

MFJL

Inscribed Circle Diameter 45 — 55 mm



Open Type



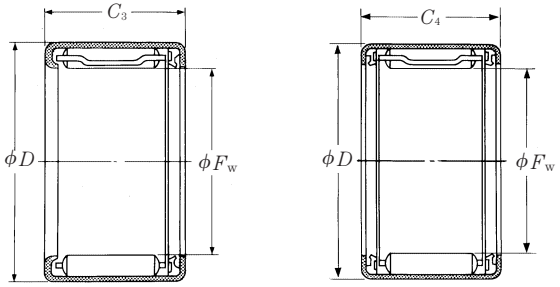
Closed Type

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min^{-1})
F_w	D	C $^{0}_{-0.25}$	C_2 max	C_r		P_{max}		Oil
45	52	16	1.0	19 900	2 030	15 400	1 570	6 700
	52	20	1.0	25 500	2 600	21 200	2 160	6 700
	52	26	1.0	34 000	3 450	30 500	3 100	6 700
	55	20	1.3	31 000	3 200	21 300	2 170	7 100
	55	25	1.3	40 500	4 150	30 000	3 050	7 100
	55	30	1.3	49 500	5 050	38 500	3 950	7 100
50	58	20	1.1	28 900	2 940	23 100	2 350	6 300
	58	24	1.1	36 000	3 700	30 500	3 150	6 300
	62	12	1.3	21 200	2 160	11 600	1 180	6 300
	62	15	1.3	24 200	2 470	13 800	1 400	6 300
	62	20	1.3	37 000	3 800	23 700	2 420	6 300
	62	25	1.3	48 500	4 950	33 500	3 450	6 300
	62	30	1.3	59 500	6 050	43 500	4 450	6 300
55	63	20	1.1	30 000	3 100	25 100	2 560	5 600
	63	24	1.1	37 500	3 850	33 500	3 400	5 600
	67	30	1.3	62 500	6 350	47 500	4 850	5 600
	67	40	1.3	81 500	8 350	67 000	6 850	5 600

Bearing Numbers				Mass (g)		Matching Inner Rings	
Open		Closed		approx			
FJ	FJL	MFJ	MFJL	Open	Closed		
FJ-4516		**MFJ-4516**		39	50	—	—
FJ-4520		**MFJ-4520**		53	59	**FIR-404520**	—
FJ-4526		**MFJ-4526**		68	74	**FIR-404526**	—
	FJL-4520L		**MFJL-4520L**	74	82	**FIR-404520**	—
	FJL-4525L		**MFJL-4525L**	94	105	**FIR-404525**	—
	FJL-4530L		**MFJL-4530L**	110	125	**FIR-404530**	—
FJ-5020L		**MFJ-5020**		56	71	**FIR-455020**	—
FJ-5024		**MFJ-5024**		69	84	—	—
	FJL-5012L		**MFJL-5012L**	60	66	**FIR-455012**	—
	FJL-5015L		**MFJL-5015L**	75	83	**FIR-455015**	—
	FJL-5020L		**MFJL-5020L**	100	110	**FIR-455020**	—
	FJL-5025L		**MFJL-5025L**	125	140	**FIR-455025**	—
	FJL-5030L		**MFJL-5030L**	150	165	**FIR-455030**	—
FJ-5520		**MFJ-5520**		60	79	—	—
FJ-5524		**MFJ-5524**		72	90	—	—
	FJL-5530L		**MFJL-5530L**	155	180	—	—
	FJL-5540L		**MFJL-5540L**	195	220	—	—

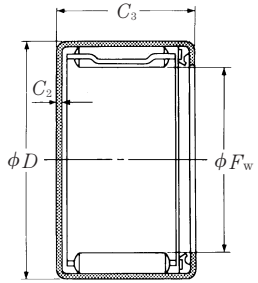
Sealed Type With Cage

FJT (**Open Type • One-Side Sealed Type**)
FJHT
FJLT
FJTT (**Open Type • Both-Side Sealed Type**)
FJHTT
FJLTT
MFJT (Closed Type • Sealed Type)
MFJLT
Inscribed Circle Diameter 8 — 22 mm



Open Type

FJT, FJHT, FJLT FJTT, FJHTT, FJLTT



Closed Type

Boundary Dimensions (mm)					Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds(1) (min^{-1})	Bearing Numbers Open	
F_w	D	C_3 $^{0}_{-0.25}$	C_4 $^{0}_{-0.25}$	C_2 max	C_r		P_{max}			FJT FJHT	FJTT FJHTT
8	12	12	14	0.8	2 840	289	1 270	130	22 000	**FJT-812**	**FJTT-814**
	15	12	14	1.0	4 400	450	1 770	181	22 000		
10	14	12	14	0.8	3 500	360	1 760	179	19 000	**FJT-1012**	**FJTT-1014**
	17	12	14	1.0	4 900	500	1 990	203	19 000		
12	16	12	14	0.8	4 150	420	2 210	225	15 000	**FJT-1212**	**FJTT-1214**
	18	14	16	1.0	6 450	655	3 050	310	15 000	**FJHT-1214**	**FJHTT-1216**
	19	14	16	1.0	6 650	675	2 950	300	15 000		
14	20	14	16	1.0	6 500	665	3 250	335	13 000	**FJT-1414**	**FJTT-1416**
	22	19	22	1.1	11 600	1 180	5 700	580	13 000		
15	21	14	16	1.0	7 650	780	3 900	400	12 000	**FJT-1514**	**FJTT-1516**
	22	13	16	1.0	6 450	655	2 880	294	12 000		
	22	18	21	1.0	10 300	1 050	5 300	540	12 000		
16	22	14	16	1.0	7 100	725	3 750	380	11 000	**FJT-1614**	**FJTT-1616**
	22	18	20	1.0	10 400	1 060	6 050	620	11 000	**FJT-1618**	**FJTT-1620**
	24	23	26	1.1	17 100	1 740	9 500	970	11 000		
17	24	18	21	1.0	10 800	1 110	5 700	580	11 000		
18	24	14	16	1.0	7 650	780	4 200	430	10 000	**FJT-1814**	**FJTT-1816**
	25	18	21	1.0	11 400	1 160	6 100	620	10 000		
	25	20	23	1.0	13 300	1 360	7 500	765	10 000		
20	26	14	16	1.0	8 150	835	4 650	475	9 500	**FJT-2014**	**FJTT-2016**
	26	18	20	1.0	11 900	1 210	7 550	770	9 500	**FJT-2018**	**FJTT-2020**
	27	18	21	1.0	12 300	1 260	6 900	705	9 500		
	27	23	26	1.0	17 500	1 790	10 900	1 110	9 500		
22	28	14	16	1.0	8 650	880	5 150	525	8 500	**FJT-2214**	**FJTT-2216**
	28	18	20	1.0	12 600	1 290	8 350	850	8 500	**FJT-2218**	**FJTT-2220**
	29	18	21	1.0	12 700	1 300	7 300	745	8 500		
	29	23	26	1.0	18 100	1 840	11 500	1 170	8 500		

Bearing Numbers Open		Closed	Mass (g) approx		
FJLT	FJLTT	MFJT MFJLT	FJT FJHT FJLT	FJTT FJLTT FJHTT	MFJT MFJLT
MFJT-812			3.1	3.5	3.6
FJLT-812	**FJLTT-814**	**MFJLT-812**	6.4	6.8	7.0
MFJT-1012			4.1	4.6	4.6
FJLT-1012	**FJLTT-1014**	**MFJLT-1012**	7.5	8.0	8.3
MFJT-1212			5.1	6.1	5.5
MFJHT-1214			8.7	9.7	9.2
FJLT-1214	**FJLTT-1216**	**MFJLT-1214**	11	12	12
MFJT-1414			11	13	12
FJLT-1419	**FJLTT-1422**	**MFJLT-1419**	18	19	20
MFJT-1514			11	12	12
FJLT-1513	**FJLTT-1516**	**MFJLT-1513**	11	12	12
FJLT-1518	**FJLTT-1521**	**MFJLT-1518**	15	16	17
MFJT-1614			12	13	13
MFJT-1618			15	16	16
FJLT-1623	**FJLTT-1626**	**MFJLT-1623**	25	16	28
FJLT-1718	**FJLTT-1721**	**MFJLT-1718**	17	18	19
MFJT-1814			13	15	15
FJLT-1818	**FJLTT-1821**	**MFJLT-1818**	18	19	20
FJLT-1820	**FJLTT-1823**	**MFJLT-1820**	21	22	22
MFJT-2014			14	17	16
MFJT-2018			18	21	20
FJLT-2018	**FJLTT-2021**	**MFJLT-2018**	20	22	22
FJLT-2023	**FJLTT-2026**	**MFJLT-2023**	25	27	28
MFJT-2214			16	18	19
MFJT-2218			21	23	24
FJLT-2218	**FJLTT-2221**	**MFJLT-2218**	22	24	24
FJLT-2223	**FJLTT-2226**	**MFJLT-2223**	29	30	32

Note (1) This is the value when the speed at the seal contact surface is 10 m/sec under inner ring rotation. Under outer ring rotation, the limiting speed is half this value.

Remarks Standard grease is packed in the sealed drawn cup needle roller bearings unless otherwise specified.

Sealed Type With Cage

FJT (**Open Type • One-Side Sealed Type**)

FJLT

FJTT (**Open Type • Both-Side Sealed Type**)

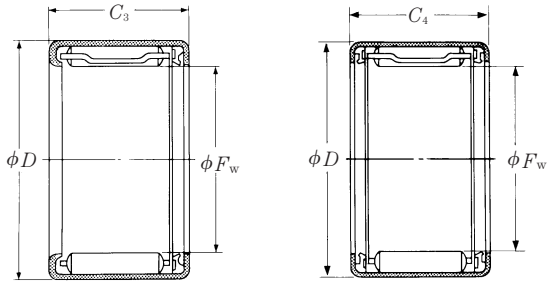
FJLTT

MFJT (Closed Type • Sealed Type)

MFJLT

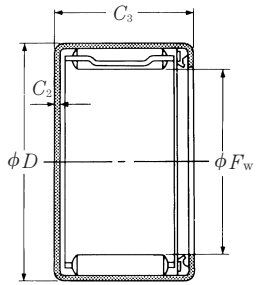
Inscribed Circle

Diameter 25 — 50 mm



Open Type

FJT, FJLT FJTT, FJLTT



Closed Type

Boundary Dimensions (mm)					Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds(1) (min^{-1})	Bearing Numbers Open		Closed	Mass (g) approx		
F_w	D	C_3 $^{0}_{-0.25}$	C_4 $^{0}_{-0.25}$	C_2 max	C_r		P_{max}			FJT	FJTT	MFJT	FJT FJLT	FJTT FJLTT	MFJT MFJLT
25	32	18	20	1.0	15 200	1 550	9 350	955	7 500	**FJT-2518**	**FJTT-2520**	**MFJT-2518**	28	30	31
	33	18	21	1.1	15 400	1 570	8 800	895	7 500			**MFJLT-2518**	28	31	31
	33	23	26	1.1	21 900	2 240	13 900	1 420	7 500			**MFJLT-2523**	37	39	41
28	35	18	20	1.0	15 600	1 590	9 950	1 020	6 700	**FJT-2818**	**FJTT-2820**	**MFJT-2818**	29	32	33
	37	23	26	1.0	25 400	2 590	15 500	1 580	6 700			**MFJLT-2823**	46	49	50
30	37	22	24	1.0	19 400	1 970	13 300	1 360	6 300	**FJT-3022**	**FJTT-3024**	**MFJT-3022**	40	42	44
	40	23	26	1.3	25 800	2 630	15 300	1 560	6 300			**MFJLT-3023**	55	57	60
	40	28	31	1.3	33 500	3 450	21 600	2 200	6 300			**MFJLT-3028**	68	70	75
35	42	22	24	1.0	23 600	2 410	17 900	1 830	5 300	**FJT-3522**	**FJTT-3524**	**MFJT-3522**	46	49	50
	45	18	21	1.3	18 200	1 860	10 100	1 030	5 300			**MFJLT-3518**	47	50	52
	45	28	31	1.3	36 000	3 650	24 000	2 450	5 300			**MFJLT-3528**	77	80	85
40	47	22	24	1.0	23 500	2 400	18 500	1 890	4 500	**FJT-4022**	**FJTT-4024**	**MFJT-4022**	52	55	57
	50	18	21	1.3	19 900	2 030	11 600	1 180	4 500			**MFJLT-4018**	54	57	59
	50	28	31	1.3	39 000	3 950	27 600	2 810	4 500			**MFJLT-4028**	88	91	97
45	52	22	24	1.0	25 500	2 600	21 200	2 160	4 000	**FJT-4522**	**FJTT-4524**	**MFJT-4522**	61	64	67
	55	23	26	1.3	31 000	3 200	21 300	2 170	4 000			**MFJLT-4523**	78	82	86
50	62	28	31	1.3	48 500	4 950	33 500	3 450	3 800			**MFJLT-5028**	130	135	145

		Open		Closed
		FJLT	FJLTT	MFJLT
MFJT-2518		**FJLT-2518**	**FJLTT-2521**	**MFJLT-2518**
		FJLT-2523	**FJLTT-2526**	**MFJLT-2523**
MFJT-2818		**FJLT-2823**	**FJLTT-2826**	**MFJLT-2823**
MFJT-3022		**FJLT-3023**	**FJLTT-3026**	**MFJLT-3023**
		FJLT-3028	**FJLTT-3031**	**MFJLT-3028**
MFJT-3522		**FJLT-3518**	**FJLTT-3521**	**MFJLT-3518**
		FJLT-3528	**FJLTT-3531**	**MFJLT-3528**
MFJT-4022		**FJLT-4018**	**FJLTT-4021**	**MFJLT-4018**
		FJLT-4028	**FJLTT-4031**	**MFJLT-4028**
MFJT-4522		**FJLT-4523**	**FJLTT-4526**	**MFJLT-4523**
		FJLT-5028	**FJLTT-5031**	**MFJLT-5028**

Note (1) This is the value when the speed at the seal contact surface is 10 m/sec under inner ring rotation. Under outer ring rotation, the limiting speed is half this value.

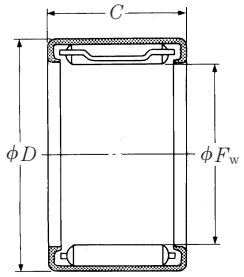
Remarks Standard grease is packed in the sealed drawn cup needle roller bearings unless otherwise specified.

With Cage

Thin Type

DB

Inscribed Circle Diameter 18 — 82 mm



Boundary Dimensions (mm)			Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	D	C $^{0}_{-0.25}$	C_r		P_{max}		Oil		approx
18	23	9.5	5 050	515	2 820	288	18 000	**DB501801**	7.5
33	39	26	22 500	2 300	19 500	1 980	9 500	**DB503303A**	44.4
39	45	12	6 500	665	4 200	430	8 000	**DB503901**	23
55	61	12	9 900	1 010	8 050	820	5 600	**DB505501**	35
82	88	12	9 400	955	8 400	855	3 600	**DB508201**	52

Full Complement Type

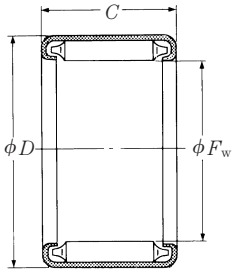
F • FH (Open Type)

NF

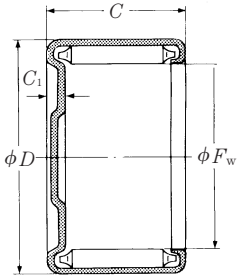
MF • MFH (Closed Type)

MNF

Inscribed Circle Diameter 8 — 20 mm



Open Type



Closed Type

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min^{-1})
F_w	D	C $^{0}_{-0.25}$	C_1 max	C_r		P_{max}		Oil
8	14	10	1.9	5 550	565	2 980	305	10 000
9	15	10	1.8	6 100	625	3 350	340	10 000
10	14	10	1.6	5 800	595	4 050	415	5 600
	16	10	1.9	6 650	680	3 700	375	9 000
12	16	10	1.6	6 450	660	4 700	480	5 000
	16	12	1.6	7 950	810	6 200	630	5 000
	18	12	1.9	9 000	920	5 700	580	7 500
13	19	12	1.9	9 550	975	6 100	625	7 100
14	20	12	2.2	9 450	965	6 350	645	6 000
	20	13	2.2	10 400	1 060	7 200	735	6 000
	20	16	2.2	13 300	1 360	9 850	1 000	6 000
	19	16	1.6	12 900	1 310	10 400	1 060	5 300
	19	20	1.6	16 100	1 640	13 800	1 410	5 300
15	21	10	1.8	8 050	820	5 000	510	6 000
	21	12	1.8	10 300	1 050	6 900	705	6 000
	21	14	1.8	12 400	1 270	8 800	895	6 000
	21	16	1.8	14 500	1 480	10 700	1 090	6 000
16	22	12	2.2	10 200	1 040	7 100	725	5 300
	22	16	2.2	14 400	1 460	11 100	1 130	5 300
17	23	12	1.8	11 300	1 150	7 750	790	5 600
	23	16	1.8	15 800	1 610	12 000	1 220	5 600
	23	20	1.8	20 200	2 060	16 300	1 670	5 600
18	24	12	2.2	10 900	1 110	7 900	805	5 000
	24	16	2.2	15 300	1 560	12 300	1 250	5 000
20	26	12	2.2	11 500	1 170	8 700	885	4 500
	26	16	2.2	16 200	1 650	13 500	1 380	4 500
	26	20	2.2	20 500	2 090	18 300	1 870	4 500

Bearing Numbers		Mass (g)		Matching Inner Rings	
Open F, FH NF	Closed MF, MFH MNF	approx Open	Closed		
FH-810	**MFH-810**	6.0	6.3	—	—
FH-910	**MFH-910**	6.4	6.8	—	—
F-1010	**MF-1010**	4.7	5.0	**FIR-071010**	—
FH-1010	**MFH-1010**	6.9	7.3	**FIR-071010**	—
F-1210	**MF-1210**	5.5	5.4	**FIR-081210**	—
F-1212	**MF-1212**	6.6	6.8	**FIR-081212**	—
FH-1212	**MFH-1212**	10	11	**FIR-081212**	—
F-1312	**MF-1312**	11	12	**FIR-101312**	—
F-1412	**MF-1412**	12	14	**FIR-101412**	—
F-1413	**MF-1413**	13	15	**FIR-101413**	—
F-1416	**MF-1416**	18	19	—	—
NF-1416	**MNF-1416**	12	13	**FIR-101416**	—
NF-1420	**MNF-1420**	16	17	**FIR-101420**	—
F-1510	**MF-1510**	10	12	**FIR-101510**	—
F-1512	**MF-1512**	12	14	**FIR-101512**	**FIR-121512**
F-1514	**MF-1514**	15	16	—	—
F-1516	**MF-1516**	17	18	**FIR-121516**	—
F-1612	**MF-1612**	14	15	**FIR-121612**	—
F-1616	**MF-1616**	18	20	**FIR-121616**	—
F-1712	**MF-1712**	14	15	—	—
F-1716	**MF-1716**	18	20	—	—
F-1720	**MF-1720**	23	24	**FIR-121720**	—
F-1812	**MF-1812**	14	16	**FIR-151812**	—
F-1816	**MF-1816**	19	22	**FIR-151816**	—
F-2012	**MF-2012**	17	19	**FIR-172012**	—
F-2016	**MF-2016**	22	25	**FIR-172016**	—
F-2020	**MF-2020**	28	30	**FIR-152020**	**FIR-172020**

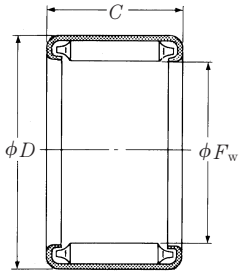
Full Complement Type

F (Open Type)

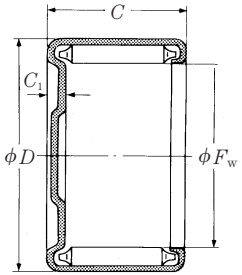
NFH

MF (Closed Type)

Inscribed Circle Diameter 22 — 55 mm



Open Type



Closed Type

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min^{-1})
F_w	D	C $^{0}_{-0.25}$	C_1 max	C_r		P_{max}		Oil
22	28	10	2.2	9 400	960	6 850	700	4 000
	28	12	2.2	12 100	1 230	9 500	970	4 000
	28	16	2.2	17 100	1 740	14 800	1 510	4 000
	28	20	2.2	21 600	2 200	20 000	2 040	4 000
24	31	25	—	31 500	3 250	28 700	2 930	4 800
25	32	16	2.5	20 200	2 060	16 200	1 650	4 500
	32	20	2.5	25 900	2 640	22 200	2 260	4 500
	32	26	2.5	34 000	3 450	31 500	3 200	4 500
28	35	12	2.5	14 800	1 510	11 200	1 150	4 000
	35	16	2.5	21 300	2 170	17 900	1 820	4 000
	35	20	2.5	27 300	2 780	24 600	2 510	4 000
	35	26	2.5	35 500	3 650	34 500	3 550	4 000
30	37	14	2.5	18 900	1 930	15 500	1 590	3 800
	37	16	2.5	22 100	2 250	18 900	1 930	3 800
	37	20	2.5	28 400	2 900	26 200	2 670	3 800
	37	26	2.5	37 000	3 800	37 000	3 750	3 800
35	42	12	2.5	16 900	1 720	14 000	1 420	3 400
	42	16	2.5	24 000	2 450	22 000	2 240	3 400
	42	20	2.5	31 000	3 150	30 000	3 100	3 400
	42	26	2.5	40 000	4 100	42 500	4 350	3 400
40	47	16	2.5	25 700	2 620	24 900	2 540	3 000
	47	20	2.5	32 500	3 350	34 000	3 450	3 000
	47	26	2.5	43 000	4 350	48 000	4 900	3 000
45	52	16	2.5	27 300	2 790	27 800	2 840	2 600
	52	20	2.5	35 000	3 550	38 500	3 900	2 600
	52	26	2.5	45 500	4 650	54 000	5 500	2 600
50	58	20	2.8	39 500	4 050	41 500	4 250	2 800
	58	24	2.8	48 000	4 900	53 000	5 400	2 800
55	63	20	2.8	41 500	4 250	45 500	4 650	2 400
	63	24	2.8	50 500	5 150	58 000	5 950	2 400

Bearing Numbers		Mass (g) approx		Matching Inner Rings	
Open	Closed				
F, NFH	MF	Open	Closed		
F-2210	**MF-2210**	14	17	**FIR-172210**	—
F-2212	**MF-2212**	18	21	**FIR-172212**	—
F-2216	**MF-2216**	24	27	**FIR-172216**	—
F-2220	**MF-2220**	30	33	**FIR-172220**	—
NFH-222A	—	45	—	—	—
F-2516	**MF-2516**	31	35	**FIR-202516**	—
F-2520	**MF-2520**	40	43	**FIR-202520**	—
F-2526	**MF-2526**	52	55	**FIR-202526**	—
F-2812	**MF-2812**	26	32	—	—
F-2816	**MF-2816**	35	40	**FIR-222816**	—
F-2820	**MF-2820**	44	48	**FIR-222820**	—
F-2826	**MF-2826**	57	62	**FIR-222826**	—
F-3014	**MF-3014**	30	34	—	—
F-3016	**MF-3016**	35	40	—	—
F-3020	**MF-3020**	46	51	**FIR-253020**	—
F-3026	**MF-3026**	61	66	**FIR-253026**	—
F-3512	**MF-3512**	31	38	**FIR-303512**	—
F-3516	**MF-3516**	42	49	—	—
F-3520	**MF-3520**	53	60	**FIR-303520**	—
F-3526	**MF-3526**	70	76	**FIR-303526**	—
F-4016	**MF-4016**	48	56	—	—
F-4020	**MF-4020**	60	69	**FIR-354020**	—
F-4026	**MF-4026**	79	88	**FIR-354026**	—
F-4516	**MF-4516**	53	64	—	—
F-4520	**MF-4520**	67	78	**FIR-404520**	—
F-4526	**MF-4526**	88	99	**FIR-404526**	—
F-5020	**MF-5020**	81	95	—	—
F-5024	**MF-5024**	98	110	—	—
F-5520	**MF-5520**	88	105	—	—
F-5524	**MF-5524**	105	125	—	—

With Cage

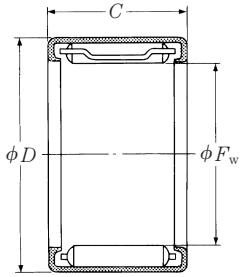
J (Open Type)

JH

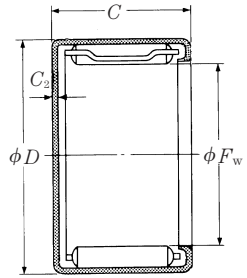
MJ (Closed Type)

MJH

Inscribed Circle Diameter 9.525 — 19.050 mm



Open Type



Closed Type

Boundary Dimensions (mm, inch) F_w		D		C $^{0}_{-0.25}$		C_2 max	Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}
9.525	0.3750	15.875	0.6250	12.70	0.5000	1.0	6 650	675	3 000	310
11.112	0.4375	15.875	0.6250	12.70	0.5000	1.0	5 800	595	3 100	315
		17.462	0.6875	12.70	0.5000	1.0	7 150	730	3 300	340
12.700	0.5000	17.462	0.6875	7.92	—	1.0	3 150	325	1 440	147
		17.462	0.6875	12.70	0.5000	1.0	6 350	650	3 500	360
		17.462	0.6875	19.05	0.7500	—	10 100	1 030	6 450	655
		19.050	0.7500	11.13	—	1.0	6 450	655	2 930	298
		19.050	0.7500	12.70	0.5000	1.0	7 600	775	3 600	370
		19.050	0.7500	19.05	0.7500	1.0	12 400	1 260	6 800	690
14.288	0.5625	19.050	0.7500	11.13	—	1.0	5 350	545	2 860	292
		19.050	0.7500	12.70	0.5000	1.0	6 250	640	3 500	360
		19.050	0.7500	15.88	0.6250	1.0	8 200	835	5 000	510
		20.638	0.8125	12.70	0.5000	1.0	8 050	820	3 950	400
		20.638	0.8125	19.05	0.7500	1.0	13 100	1 330	7 350	750
15.875	0.6250	20.638	0.8125	12.70	0.5000	1.0	6 750	685	3 950	405
		20.638	0.8125	15.88	0.6250	1.0	8 850	900	5 600	570
		20.638	0.8125	19.05	0.7500	1.0	11 600	1 190	8 050	820
		22.225	0.8750	15.88	0.6250	1.0	11 600	1 190	6 400	655
		22.225	0.8750	25.40	1.0000	1.0	19 900	2 030	12 800	1 300
17.462	0.6875	22.225	0.8750	19.05	0.7500	1.0	11 500	1 170	8 050	820
		23.812	0.9375	15.88	0.6250	1.0	12 100	1 230	6 850	700
		23.812	0.9375	19.05	0.7500	1.0	15 000	1 530	9 000	915
19.050	0.7500	25.400	1.0000	9.52	0.3750	—	5 950	610	2 800	285
		25.400	1.0000	12.70	0.5000	—	10 000	1 020	5 450	555
		25.400	1.0000	15.88	0.6250	1.0	12 600	1 280	7 250	740
		25.400	1.0000	19.05	0.7500	1.0	15 600	1 590	9 600	980
		26.988	1.0625	19.05	0.7500	1.0	19 200	1 950	10 900	1 110

Remarks Sealed bearings are available upon the request.

Limiting Speeds (min^{-1}) Oil	Bearing Numbers Open J	Open JH	Closed MJ	Closed MJH	Mass (g) approx Open	Mass (g) approx Closed	Matching Inner Rings	
38 000		**JH-68**		**MJH-681**	8.2	8.6	—	—
30 000	**J-78**		**MJ-781**		6.4	7.3	—	—
30 000		**JH-78**		**MJH-781**	9.1	9.5	—	—
26 000	**J-85**		**MJ-851**		5.4	6.4	—	—
26 000	**J-88**		**MJ-881**		7.3	8.2	—	—
26 000	**J-812**		—		7.7	—	—	—
26 000		**JH-87**		**MJH-871**	9.5	11	—	—
26 000		**JH-88**		**MJH-881**	10	13	—	—
26 000		**JH-812**		**MJH-8121**	15	17	—	—
22 000	**J-97**		**MJ-971**		7.3	8.6	—	—
22 000	**J-98**		**MJ-981**		8.2	9.5	**IR-68**	—
22 000	***J-910-F**		***MJ-9101-F**		9.1	10	—	—
24 000		**JH-98**		**MJH-98**	11	13	**IR-68**	—
24 000		***JH-912-F**		***MJH-9121-F**	14	16	—	—
20 000	**J-108**		**MJ-1081**		8.6	10	**IR-68-1**	—
20 000	**J-1010**		**MJ-10101**		10	12	—	—
20 000	**J-1012**		**MJ-10121**		13	14	**IR-612-1**	—
20 000		**JH-1010**		**MJH-10101**	15	18	—	—
20 000		**JH-1016**		**MJH-10161**	25	27	—	—
18 000	**J-1112**		**MJ-11121**		16	18	—	—
19 000		**JH-1110**		**MJH-11101**	16	18	—	—
19 000		**JH-1112**		**MJH-11121**	17	20	—	—
17 000	**J-126**		—		8.2	—	—	—
17 000	**J-128**		—		12	—	**IR-88**	—
17 000	**J-1210**		**MJ-12101**		16	19	—	—
17 000	**J-1212**		**MJ-12121**		20	20	**IR-812**	—
18 000		**JH-1212**		**MJH-12121**	24	26	**IR-812**	—

* These bearings have polyamide cages. The maximum permissible operating temperature for these bearings is 100°C for continued operation and 120°C for short periods.

DRAWN CUP NEEDLE ROLLER BEARINGS (INCH DESIGN)

With Cage

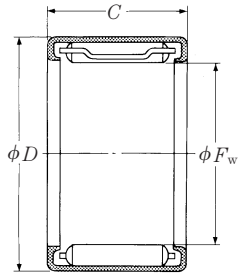
J (Open Type)

JH

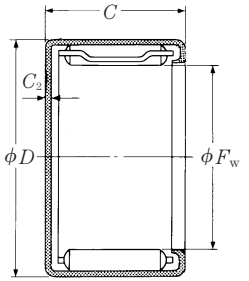
MJ (Closed Type)

MJH

Inscribed Circle Diameter 20.638 — 50.800 mm



Open Type



Closed Type

Boundary Dimensions (mm, inch) F_w		D		C $^{0}_{-0.25}$		C_2 max	Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}
20.638	0.8125	26.988	1.0625	22.22	0.8750	—	19 400	1 980	12 900	1 320
		28.575	1.1250	19.05	0.7500	1.3	18 900	1 930	11 100	1 130
22.225	0.8750	28.575	1.1250	9.52	0.3750	—	6 600	675	3 300	335
		28.575	1.1250	12.70	0.5000	—	11 000	1 120	6 350	645
		28.575	1.1250	19.05	0.7500	—	17 900	1 830	11 900	1 210
		28.575	1.1250	25.40	1.0000	1.0	23 800	2 420	17 000	1 730
		30.162	1.1875	19.05	0.7500	1.3	18 400	1 880	11 000	1 120
		30.162	1.1875	25.40	1.0000	1.3	25 500	2 600	16 700	1 700
25.400	1.0000	31.750	1.2500	19.05	0.7500	—	18 200	1 860	12 400	1 270
		31.750	1.2500	25.40	1.0000	1.0	25 000	2 550	18 700	1 910
		33.338	1.3125	19.05	0.7500	1.3	20 800	2 120	13 000	1 330
		33.338	1.3125	25.40	1.0000	1.3	27 700	2 830	18 900	1 930
28.575	1.1250	34.925	1.3750	12.70	0.5000	1.0	11 700	1 200	7 250	740
		34.925	1.3750	19.05	0.7500	1.0	19 100	1 950	13 600	1 380
		34.925	1.3750	25.40	1.0000	1.0	26 200	2 670	20 400	2 080
		38.100	1.5000	19.05	0.7500	1.3	23 500	2 390	14 000	1 420
		38.100	1.5000	25.40	1.0000	1.3	32 000	3 250	20 700	2 110
		38.100	1.5000	28.58	1.1250	1.3	36 500	3 700	24 600	2 510
31.750	1.2500	38.100	1.5000	19.05	0.7500	1.0	19 900	2 030	14 700	1 500
		38.100	1.5000	25.40	1.0000	1.0	27 400	2 790	22 100	2 250
		41.275	1.6250	19.05	0.7500	—	24 200	2 470	14 900	1 520
		41.275	1.6250	25.40	1.0000	1.3	34 000	3 500	23 200	2 360
		41.275	1.6250	31.75	1.2500	—	42 500	4 350	31 000	3 150
34.925	1.3750	41.275	1.6250	12.70	0.5000	1.0	12 400	1 260	8 150	830
		44.450	1.7500	19.05	0.7500	1.3	26 400	2 690	16 800	1 710
		44.450	1.7500	25.40	1.0000	1.3	36 500	3 750	25 600	2 610
38.100	1.5000	47.625	1.8750	19.05	0.7500	1.3	28 500	2 900	19 000	1 930
		47.625	1.8750	25.40	1.0000	1.3	39 500	4 000	28 900	2 950
		47.625	1.8750	31.75	1.2500	1.3	49 500	5 050	39 000	3 950
41.275	1.6250	50.800	2.0000	25.40	1.0000	1.3	39 500	4 000	29 500	3 000
44.450	1.7500	53.975	2.1250	19.05	0.7500	1.3	29 700	3 050	21 000	2 140
		53.975	2.1250	25.40	1.0000	1.3	40 000	4 100	31 000	3 150
		53.975	2.1250	38.10	1.5000	1.3	60 000	6 100	52 000	5 300
47.625	1.8750	57.150	2.2500	25.40	1.0000	1.3	41 500	4 200	32 000	3 300
50.800	2.0000	60.325	2.3750	25.40	1.0000	1.3	42 500	4 350	34 500	3 500

Limiting Speeds (min^{-1})	Bearing Numbers				Mass (g)		Matching Inner Rings	
	Open		Closed		approx			
Oil	J	JH	MJ	MJH	Open	Closed		
16 000	**J-1314**		—		25	—	—	—
16 000		**JH-1312**		**MJH-13121**	29	31	—	—
14 000	**J-146**		—		11	—	—	—
14 000	**J-148**		—		16	—	—	—
14 000	**J-1412**		—		25	—	**IR-1012**	—
14 000	**J-1416**		**MJ-14161**		33	36	**IR-1016**	—
15 000		**JH-1412**		**MJH-14121**	31	36	**IR-1012**	—
15 000		**JH-1416**		**MJH-14161**	42	47	**IR-1016**	—
12 000	**J-1612**		—		28	—	**IR-1212**	**IR-1312**
12 000	**J-1616**		**MJ-16161**		36	40	**IR-1216**	**IR-1316**
13 000		**JH-1612**		**MJH-16121**	34	40	**IR-1212**	**IR-1312**
13 000		**JH-1616**		**MJH-16161**	46	51	**IR-1216**	**IR-1316**
11 000	**J-188**		**MJ-1881**		27	30	—	—
11 000	**J-1812**		**MJ-18121**		30	35	—	—
11 000	**J-1816**		**MJ-18161**		47	51	**IR-1416**	**IR-1516**
11 000		**JH-1812**		**MJH-18121**	45	52	—	—
11 000		**JH-1816**		**MJH-18161**	62	69	**IR-1416**	**IR-1516**
11 000		**JH-1818**		**MJH-18181**	71	78	—	—
10 000	**J-2012**		**MJ-20121**		43	46	**IR-1612**	—
10 000	**J-2016**		**MJ-20161**		61	65	**IR-1616**	—
10 000		**JH-2012**		—	66	—	—	—
10 000		**JH-2016**		**MJH-20161**	70	76	**IR-1616**	—
10 000		**JH-2020**		—	75	—	—	—
9 000	**J-228**		**MJ-2281**		68	77	—	—
9 000		**JH-2212**		**MJH-22121**	53	61	**IR-1812**	—
9 000		**JH-2216**		**MJH-22161**	73	81	**IR-1816**	—
8 500	**J-2412**		**MJ-24121**		58	68	—	—
8 500	**J-2416**		**MJ-24161**		81	90	**IR-1916**	**IR-2016**
8 500	**J-2420**		**MJ-24201**		105	115	**IR-1920**	**IR-2020**
7 500	**J-2616**		**MJ-26161**		89	96	—	—
7 100	**J-2812**		**MJ-28121**		91	105	—	—
7 100	**J-2816**		**MJ-28161**		92	105	**IR-2316**	**IR-2426**
7 100	**J-2824**		**MJ-28241**		140	155	**IR-2324**	**IR-2424**
6 700	**J-3016**		**MJ-30161**		96	135	—	—
6 300	**J-3216**		**MJ-32161**		99	165	—	—

Remarks Sealed bearings are available upon the request.

Full Complement Type

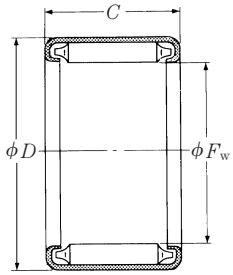
B (Open Type)

BH

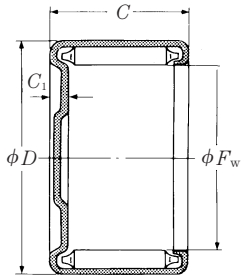
M (Closed Type)

MH

Inscribed Circle Diameter 11.112 — 15.875 mm



Open Type



Closed Type

Boundary Dimensions (mm, inch) F_w		D		C $^{0}_{-0.25}$		C_1 max	Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}
11.112	0.4375	15.875	0.6250	9.52	0.3750	—	6 050	620	3 900	400
		15.875	0.6250	11.13	—	—	7 350	750	5 000	510
		15.875	0.6250	12.70	0.5000	2.0	8 600	875	6 100	625
		15.875	0.6250	15.88	0.6250	—	10 900	1 110	8 300	850
		17.462	0.6875	12.70	0.5000	2.3	9 850	1 000	6 000	610
12.700	0.5000	17.462	0.6875	7.92	—	2.0	5 100	520	3 200	325
		17.462	0.6875	9.52	0.3750	2.0	6 550	665	4 400	450
		17.462	0.6875	11.13	—	2.0	7 950	810	5 650	575
		17.462	0.6875	12.70	0.5000	2.0	9 250	945	6 900	700
		17.462	0.6875	15.88	0.6250	2.0	11 800	1 200	9 350	955
		17.462	0.6875	19.05	0.7500	2.0	14 100	1 440	11 800	1 210
		19.050	0.7500	11.13	—	—	9 000	920	5 400	550
		19.050	0.7500	12.70	0.5000	2.3	10 700	1 090	6 700	685
		19.050	0.7500	15.88	0.6250	2.3	13 800	1 400	9 350	950
		19.050	0.7500	19.05	0.7500	2.3	16 700	1 700	11 900	1 220
14.288	0.5625	19.050	0.7500	7.92	—	2.0	5 400	550	3 550	360
		19.050	0.7500	9.52	0.3750	2.0	7 000	710	4 900	500
		19.050	0.7500	11.13	—	2.0	8 500	865	6 300	640
		19.050	0.7500	12.70	0.5000	2.0	9 900	1 010	7 650	780
		19.050	0.7500	15.88	0.6250	2.0	12 500	1 280	10 400	1 060
		19.050	0.7500	19.05	0.7500	2.0	15 100	1 540	13 200	1 340
		20.638	0.8125	12.70	0.5000	—	11 400	1 170	7 400	755
		20.638	0.8125	15.88	0.6250	—	14 800	1 510	10 300	1 050
		20.638	0.8125	19.05	0.7500	2.3	17 900	1 830	13 200	1 350
15.875	0.6250	20.638	0.8125	7.92	—	2.0	5 750	585	3 900	395
		20.638	0.8125	11.13	—	2.0	8 950	915	6 900	705
		20.638	0.8125	12.70	0.5000	2.0	10 500	1 070	8 400	860
		20.638	0.8125	15.88	0.6250	2.0	13 300	1 350	11 400	1 170
		20.638	0.8125	19.05	0.7500	2.0	16 000	1 630	14 500	1 480
		22.225	0.8750	12.70	0.5000	2.3	12 200	1 240	8 150	830
		22.225	0.8750	15.88	0.6250	—	15 700	1 600	11 300	1 150
		22.225	0.8750	19.05	0.7500	2.3	19 000	1 940	14 500	1 470
		22.225	0.8750	25.40	1.0000	—	25 200	2 570	20 800	2 120

Limiting Speeds (min^{-1}) Oil	Bearing Numbers Open B	Open BH	Closed M	Closed MH	Mass (g) approx Open	Mass (g) approx Closed	Matching Inner Rings	
6 000	**B-76**		—		5.4	—	—	—
6 000	**B-77**		—		6.4	—	—	—
6 000	**B-78**		**M-781**		7.3	8.2	—	—
6 000	**B-710**		—		9.5	—	—	—
8 500		**BH-78**		**MH-781**	10	11	—	—
5 300	**B-85**		**M-851**		5.4	6.4	—	—
5 300	**B-86**		**M-861**		6.4	7.3	—	—
5 300	**B-87**		**M-871**		7.3	8.2	—	—
5 300	**B-88**		**M-881**		8.2	9.1	—	—
5 300	**B-810**		**M-8101**		11	12	—	—
5 300	**B-812**		**M-8121**		13	14	—	—
7 500		**BH-87**		—	7.3	—	—	—
7 500		**BH-88**		**MH-881**	11	13	—	—
7 500		**BH-810**		**MH-8101**	14	16	—	—
7 500		**BH-812**		**MH-8121**	18	19	—	—
5 000	**B-95**		**M-951**		5.7	7.1	—	—
5 000	**B-96**		**M-961**		6.4	7.7	—	—
5 000	**B-97**		**M-971**		7.7	9.1	—	—
5 000	**B-98**		**M-981**		9.1	10	**IR-68**	—
5 000	**B-910**		**M-9101**		12	13	—	—
5 000	**B-912**		**M-9121**		14	16	**IR-612**	—
7 100		**BH-98**		—	12	—	**IR-68**	—
7 100		**BH-910**		—	16	—	—	—
7 100		**BH-912**		**MH-9121**	20	21	**IR-612**	—
4 500	**B-105**		**M-1051**		5.9	7.7	—	—
4 500	**B-107**		**M-1071**		8.6	11	—	—
4 500	**B-108**		**M-1081**		10	12	**IR-68-1**	—
4 500	**B-1010**		**M-10101**		13	15	—	—
4 500	**B-1012**		**M-10121**		16	18	**IR-612-1**	—
6 300		**BH-108**		**MH-1081**	13	15	**IR-68-1**	—
6 300		**BH-1010**		—	17	—	—	—
6 300		**BH-1012**		**MH-10121**	21	23	**IR-612-1**	—
6 300		**BH-1016**		—	30	—	—	—

Full Complement Type

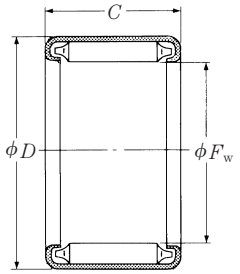
B (Open Type)

BH

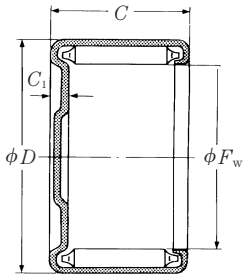
M (Closed Type)

MH

Inscribed Circle
Diameter 17.462 — 23.812 mm



Open Type



Closed Type

Boundary Dimensions (mm, inch)							Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}
F_w		D		C $^{0}_{-0.25}$		C_1 max	C_r		P_{max}	
17.462	0.6875	22.225	0.8750	9.52	0.3750	2.0	7 800	795	5 850	600
		22.225	0.8750	12.70	0.5000	2.0	11 000	1 120	9 200	935
		22.225	0.8750	15.88	0.6250	2.0	14 000	1 430	12 500	1 270
		22.225	0.8750	19.05	0.7500	2.0	16 800	1 710	15 800	1 610
		23.812	0.9375	11.13	—	—	10 800	1 100	7 100	725
		23.812	0.9375	12.70	0.5000	—	12 800	1 310	8 850	900
		23.812	0.9375	15.88	0.6250	2.3	16 600	1 690	12 300	1 250
		23.812	0.9375	19.05	0.7500	—	20 100	2 050	15 700	1 600
19.050	0.7500	25.400	1.0000	9.52	0.3750	2.3	9 150	935	5 800	595
		25.400	1.0000	12.70	0.5000	2.3	13 500	1 370	9 550	975
		25.400	1.0000	15.88	0.6250	2.3	17 400	1 770	13 300	1 350
		25.400	1.0000	19.05	0.7500	2.3	21 100	2 150	17 000	1 730
20.638	0.8125	26.988	1.0625	9.52	0.3750	—	9 600	975	6 250	635
		26.988	1.0625	12.70	0.5000	2.3	14 100	1 440	10 200	1 050
		26.988	1.0625	22.22	0.8750	2.3	25 700	2 620	22 200	2 270
		26.988	1.0625	25.40	1.0000	2.3	29 200	2 980	26 200	2 670
		28.575	1.1250	12.70	0.5000	2.8	14 100	1 440	9 100	930
		28.575	1.1250	15.88	0.6250	2.8	18 800	1 920	13 200	1 350
		28.575	1.1250	19.05	0.7500	2.8	23 200	2 360	17 300	1 760
22.225	0.8750	28.575	1.1250	9.52	0.3750	2.3	10 000	1 020	6 700	680
		28.575	1.1250	12.70	0.5000	2.3	14 700	1 490	11 000	1 120
		28.575	1.1250	19.05	0.7500	2.3	22 900	2 340	19 500	1 990
		28.575	1.1250	25.40	1.0000	2.3	30 500	3 100	28 000	2 860
		28.575	1.1250	28.58	1.1250	—	34 000	3 450	32 500	3 300
		30.162	1.1875	15.88	0.6250	2.8	19 400	1 980	14 100	1 430
		30.162	1.1875	19.05	0.7500	2.8	24 000	2 440	18 400	1 880
		30.162	1.1875	25.40	1.0000	2.8	32 500	3 300	27 100	2 770
23.812	0.9375	30.162	1.1875	12.70	0.5000	—	15 200	1 550	11 700	1 190
		30.162	1.1875	25.40	1.0000	2.3	31 500	3 200	29 800	3 050

Limiting Speeds (min^{-1})	Bearing Numbers				Mass (g)		Matching Inner Rings	
	Open		Closed		approx			
Oil	B	BH	M	MH	Open	Closed		
4 000	**B-116**		**M-1161**		10	12	—	—
4 000	**B-118**		**M-1181**		11	14	—	—
4 000	**B-1110**		**M-11101**		14	17	—	—
4 000	**B-1112**		**M-11121**		17	20	—	—
6 000		**BH-117**		—	13	—	—	—
6 000		**BH-118**		—	17	—	—	—
6 000		**BH-1110**		**MH-11101**	20	23	—	—
6 000		**BH-1112**		—	25	—	—	—
5 300	**B-126**		**M-1261**		10	13	—	—
5 300	**B-128**		**M-1281**		15	18	**IR-88**	—
5 300	**B-1210**		**M-12101**		20	23	—	—
5 300	**B-1212**		**M-12121**		25	28	**IR-812**	—
5 000	**B-136**		—		12	—	—	—
5 000	**B-138**		**M-1381**		17	20	—	—
5 000	**B-1314**		**M-13141**		32	36	—	—
5 000	**B-1316**		**M-13161**		37	40	—	—
6 000		**BH-138**		**MH-1381**	22	24	—	—
6 000		**BH-1310**		**MH-13101**	26	29	—	—
6 000		**BH-1312**		**MH-13121**	35	37	—	—
4 800	**B-146**		**M-1461**		13	17	—	—
4 800	**B-148**		**M-1481**		18	22	—	—
4 800	**B-1412**		**M-14121**		29	33	**IR-1012**	—
4 800	**B-1416**		**M-14161**		40	43	**IR-1016**	—
4 800	**B-1418**		—		50	—	—	—
5 600		**BH-1410**		**MH-14101**	30	35	—	—
5 600		**BH-1412**		**MH-14121**	36	41	**IR-1012**	—
5 600		**BH-1416**		**MH-14161**	40	43	—	—
4 500	**B-158**		—		20	—	—	—
4 500	**B-1516**		**M-15161**		41	46	—	—

Full Complement Type

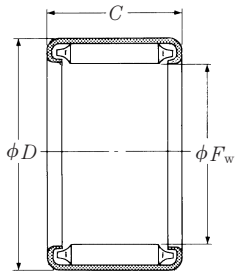
B (Open Type)

BH

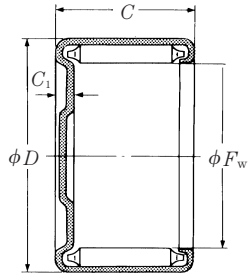
M (Closed Type)

MH

Inscribed Circle
Diameter 25.400 — 31.750 mm



Open Type



Closed Type

Boundary Dimensions (mm, inch)							Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}
F_w		D		C $^{0}_{-0.25}$		C_1 max	C_r		P_{max}	
25.400	1.0000	31.750	1.2500	9.52	0.3750	—	10 700	1 090	7 550	770
		31.750	1.2500	11.13	—	2.3	13 300	1 350	9 950	1 010
		31.750	1.2500	12.70	0.5000	2.3	15 700	1 610	12 400	1 260
		31.750	1.2500	15.88	0.6250	2.3	20 300	2 070	17 200	1 750
		31.750	1.2500	19.05	0.7500	2.3	24 700	2 510	22 000	2 240
		31.750	1.2500	25.40	1.0000	2.3	32 500	3 350	31 500	3 250
		33.338	1.3125	12.70	0.5000	2.8	15 800	1 610	10 900	1 120
		33.338	1.3125	15.88	0.6250	—	21 000	2 140	15 900	1 620
		33.338	1.3125	19.05	0.7500	2.8	25 900	2 650	20 800	2 120
		33.338	1.3125	22.22	0.8750	—	30 500	3 100	25 700	2 620
		33.338	1.3125	25.40	1.0000	2.8	35 000	3 600	30 500	3 100
		33.338	1.3125	31.75	1.2500	—	43 500	4 450	40 500	4 100
		33.338	1.3125	38.10	1.5000	2.8	51 500	5 250	50 500	5 150
26.988	1.0625	33.338	1.3125	15.88	0.6250	2.3	21 000	2 140	18 200	1 850
28.575	1.1250	34.925	1.3750	9.52	0.3750	2.3	11 400	1 160	8 400	855
		34.925	1.3750	12.70	0.5000	2.3	16 800	1 710	13 800	1 410
		34.925	1.3750	19.05	0.7500	2.3	26 200	2 680	24 500	2 500
		34.925	1.3750	25.40	1.0000	2.3	35 000	3 550	35 500	3 600
		38.100	1.5000	19.05	0.7500	3.0	30 000	3 050	22 900	2 340
		38.100	1.5000	25.40	1.0000	3.0	41 000	4 150	34 000	3 500
		38.100	1.5000	28.58	1.1250	—	46 000	4 700	39 500	4 050
		38.100	1.5000	31.75	1.2500	3.0	51 000	5 200	45 500	4 600
30.162	1.1875	38.100	1.5000	15.88	0.6250	2.8	23 000	2 350	18 500	1 890
		38.100	1.5000	25.40	1.0000	—	38 500	3 900	35 500	3 650
31.750	1.2500	38.100	1.5000	12.70	0.5000	2.3	17 700	1 810	15 200	1 550
		38.100	1.5000	15.88	0.6250	2.3	22 900	2 330	21 100	2 150
		38.100	1.5000	19.05	0.7500	2.3	27 700	2 830	27 000	2 760
		38.100	1.5000	25.40	1.0000	2.3	37 000	3 750	39 000	3 950
		38.100	1.5000	31.75	1.2500	2.3	45 500	4 600	50 500	5 150
		41.275	1.6250	12.70	0.5000	3.0	18 400	1 880	12 500	1 280
		41.275	1.6250	19.05	0.7500	3.0	31 500	3 200	25 000	2 550
		41.275	1.6250	25.40	1.0000	3.0	43 000	4 400	37 500	3 800
		41.275	1.6250	31.75	1.2500	3.0	53 500	5 450	49 500	5 050

Limiting Speeds (min^{-1})	Bearing Numbers				Mass (g)		Matching Inner Rings	
	Open		Closed		approx			
Oil	B	BH	M	MH	Open	Closed		
4 000	**B-166**		—		14	—	—	—
4 000	**B-167**		**M-1671**		17	22	—	—
4 000	**B-168**		**M-1681**		20	25	**IR-128**	—
4 000	**B-1610**		**M-16101**		26	31	—	—
4 000	**B-1612**		**M-16121**		32	37	**IR-1212**	**IR-1312**
4 000	**B-1616**		**M-16161**		44	49	**IR-1216**	**IR-1316**
5 000		**BH-168**		**MH-1681**	25	30	**IR-128**	—
5 000		**BH-1610**		—	33	—	—	—
5 000		**BH-1612**		**MH-16121**	40	45	**IR-1212**	**IR-1312**
5 000		**BH-1614**		—	48	—	—	—
5 000		**BH-1616**		**MH-16161**	55	60	**IR-1216**	**IR-1316**
5 000		**BH-1620**		—	70	—	**IR-1220**	—
5 000		**BH-1624**		**MH-16241**	85	91	**IR-1224**	—
4 000	**B-1710**		**M-17101**		27	32	—	—
3 800	**B-186**		**M-1861**		15	20	—	—
3 800	**B-188**		**M-1881**		23	28	—	—
3 800	**B-1812**		**M-18121**		35	40	—	—
3 800	**B-1816**		**M-18161**		48	54	**IR-1416**	**IR-1516**
5 300		**BH-1812**		**MH-18121**	55	62	—	—
5 300		**BH-1816**		**MH-18161**	75	82	**IR-1416**	**IR-1516**
5 300		**BH-1818**		—	86	—	—	—
5 300		**BH-1820**		**MH-18201**	98	105	—	—
4 300	**B-1910**		**M-19101**		35	41	—	—
4 300	**B-1916**		—		62	—	—	—
3 400	**B-208**		**M-2081**		24	31	—	—
3 400	**B-2010**		**M-20101**		32	38	—	—
3 400	**B-2012**		**M-20121**		39	45	**IR-1612**	—
3 400	**B-2016**		**M-20161**		54	60	**IR-1616**	—
3 400	**B-2020**		**M-20201**		68	74	—	—
5 000		**BH-208**		**MH-2081**	28	32	—	—
5 000		**BH-2012**		**MH-20121**	57	62	**IR-1612**	—
5 000		**BH-2016**		**MH-20161**	80	86	**IR-1616**	—
5 000		**BH-2020**		**MH-20201**	105	110	—	—

Full Complement Type

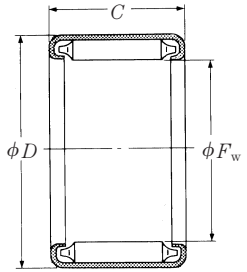
B (Open Type)

BH

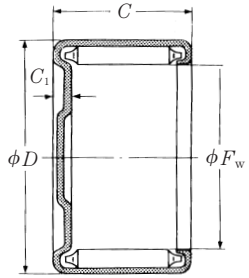
M (Closed Type)

MH

Inscribed Circle
Diameter 33.338 — 50.800 mm



Open Type



Closed Type

Boundary Dimensions (mm, inch)							Basic Dynamic Load Ratings (N) {kgf}		Limiting Loads (N) {kgf}	
F_w		D		C $^{0}_{-0.25}$		C_1 max	C_r		P_{max}	
33.338	1.3125	41.275	1.6250	12.70	0.5000	2.8	18 300	1 860	14 000	1 430
		41.275	1.6250	15.88	0.6250	2.8	24 400	2 490	20 300	2 070
34.925	1.3750	41.275	1.6250	12.70	0.5000	2.3	18 600	1 900	16 600	1 690
		41.275	1.6250	19.05	0.7500	2.3	29 200	2 970	29 500	3 000
		41.275	1.6250	25.40	1.0000	2.3	38 500	3 950	42 500	4 350
		41.275	1.6250	31.75	1.2500	2.3	47 500	4 850	55 500	5 650
		44.450	1.7500	12.70	0.5000	—	20 100	2 050	14 100	1 440
		44.450	1.7500	15.88	0.6250	—	27 200	2 770	20 800	2 120
		44.450	1.7500	19.05	0.7500	3.0	33 500	3 450	27 500	2 800
		44.450	1.7500	25.40	1.0000	3.0	46 000	4 700	41 000	4 150
		44.450	1.7500	31.75	1.2500	—	57 000	5 850	54 000	5 550
38.100	1.5000	47.625	1.8750	12.70	0.5000	3.0	20 900	2 130	15 200	1 550
		47.625	1.8750	15.88	0.6250	3.0	38 200	2 880	22 500	2 290
		47.625	1.8750	19.05	0.7500	3.0	35 000	3 600	29 700	3 050
		47.625	1.8750	22.22	0.8750	3.0	41 500	4 250	37 000	3 750
		47.625	1.8750	25.40	1.0000	3.0	48 000	4 850	44 000	4 500
		47.625	1.8750	31.75	1.2500	3.0	59 500	6 050	58 500	6 000
41.275	1.6250	50.800	2.0000	12.70	0.5000	—	21 500	2 200	16 200	1 660
		50.800	2.0000	15.88	0.6250	3.0	29 200	2 980	24 000	2 450
		50.800	2.0000	25.40	1.0000	—	49 500	5 050	47 500	4 850
		50.800	2.0000	31.75	1.2500	3.0	61 500	6 300	63 000	6 400
44.450	1.7500	53.975	2.1250	19.05	0.7500	3.0	37 500	3 850	34 000	3 450
		53.975	2.1250	25.40	1.0000	3.0	51 000	5 250	50 500	5 150
		53.975	2.1250	31.75	1.2500	—	64 000	6 500	67 500	6 850
		53.975	2.1250	38.10	1.5000	3.0	76 000	7 750	84 000	8 550
47.625	1.8750	57.150	2.2500	12.70	0.5000	3.0	23 500	2 400	18 700	1 910
		57.150	2.2500	19.05	0.7500	—	39 500	4 050	36 500	3 700
		57.150	2.2500	25.40	1.0000	3.0	54 000	5 500	54 000	5 550
50.800	2.0000	60.325	2.3750	12.70	0.5000	3.0	24 000	2 450	19 700	2 010
		60.325	2.3750	25.40	1.0000	3.0	55 000	5 650	57 500	5 850
		60.325	2.3750	31.75	1.2500	3.0	69 000	7 000	76 000	7 750
		60.325	2.3750	38.10	1.5000	3.0	82 000	8 350	95 000	9 700
		60.325	2.3750	44.45	1.7500	3.0	94 000	9 600	114 000	11 600

Limiting Speeds (min^{-1})	Bearing Numbers				Mass (g)		Matching Inner Rings	
	Open		Closed		approx			
Oil	B	BH	M	MH	Open	Closed		
4 000	**B-218**		**M-2181**		32	38	—	—
4 000	**B-2110**		**M-21101**		40	46	—	—
3 000	**B-228**		**M-2281**		27	34	—	—
3 000	**B-2212**		**M-22121**		43	49	**IR-1812**	—
3 000	**B-2216**		**M-22161**		58	65	**IR-1816**	—
3 000	**B-2220**		**M-22201**		74	83	**IR-1820**	—
4 500		**BH-228**		—	29	—	—	—
4 500		**BH-2210**		—	43	—	—	—
4 500		**BH-2212**		**MH-22121**	58	66	**IR-1812**	—
4 500		**BH-2216**		**MH-22161**	87	95	**IR-1816**	—
4 500		**BH-2220**		—	115	—	**IR-1820**	—
4 000	**B-248**		**M-2481**		39	48	—	—
4 000	**B-2410**		**M-24101**		54	63	—	—
4 000	**B-2412**		**M-24121**		69	78	—	—
4 000	**B-2414**		**M-24141**		81	90	—	—
4 000	**B-2416**		**M-24161**		92	100	**IR-1916**	**IR-2016**
4 000	**B-2420**		**M-24201**		120	130	**IR-1920**	**IR-2020**
3 800	**B-268**		—		40	—	—	—
3 800	**B-2610**		**M-26101**		57	74	—	—
3 800	**B-2616**		—		89	—	—	—
3 800	**B-2620**		**M-26201**		140	155	**IR-2220**	—
3 600	**B-2812**		**M-28121**		78	92	—	—
3 600	**B-2816**		**M-28161**		110	125	**IR-2316**	**IR-2416**
3 600	**B-2820**		—		140	—	—	—
3 600	**B-2824**		**M-28241**		170	190	**IR-2324**	**IR-2424**
3 400	**B-308**		**M-3081**		52	73	—	—
3 400	**B-3012**		—		82	—	—	—
3 400	**B-3016**		**M-30161**		110	130	—	—
3 200	**B-328**		**M-3281**		51	72	—	—
3 200	**B-3216**		**M-32161**		120	140	—	—
3 200	**B-3220**		**M-32201**		150	175	—	—
3 200	**B-3224**		**M-32241**		185	205	—	—
3 200	**B-3228**		**M-32281**		220	240	—	—

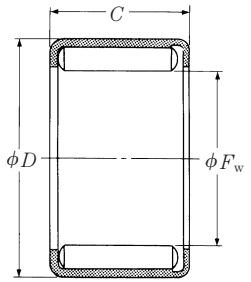
DRAWN CUP NEEDLE ROLLER BEARINGS (INCH DESIGN)

Full Complement Type (Grease-Retained Roller Type)

Y

YH

Inscribed Circle Diameter 3.969 — 20.638 mm



Boundary Dimensions (mm, inch) F_w		D		C $^{0}_{-0.25}$		Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}	Limiting Speeds (min^{-1}) Oil	Bearing Numbers	Mass (g) approx
3.969	—	7.144	—	3.96	—	1 260	129	580	59	10 000	**Y-2** 1/2**2** 1/2	0.4
		7.144	—	6.35	0.2500	2 300	235	1 260	128	10 000	**Y-2** 1/2**4**	1.4
4.762	0.1875	8.731	—	6.35	0.2500	2 560	261	1 350	137	10 000	**Y-34**	2.3
6.350	0.2500	11.112	0.4375	7.92	—	4 150	420	2 300	234	10 000	**Y-45**	3.2
		11.112	0.4375	11.13	—	5 950	605	3 700	375	10 000	**Y-47**	4.5
7.938	0.3125	12.700	0.5000	7.92	—	4 750	485	2 760	281	8 000	**Y-55**	4.1
9.525	0.3750	14.288	0.5625	6.35	0.2500	4 000	410	2 250	229	7 100	**Y-64**	3.2
		14.288	0.5625	9.52	0.3750	6 500	660	4 200	425	7 100	**Y-66**	5.5
		14.288	0.5625	12.70	0.5000	8 750	890	6 100	625	7 100	**Y-68**	7.3
		14.288	0.5625	15.88	0.6250	10 800	1 100	8 050	820	7 100	**Y-610**	9.1
11.112	0.4375	15.875	0.6250	12.70	0.5000	9 550	975	7 000	715	6 000	**Y-78**	8.2
		17.462	0.6875	12.70	0.5000	11 400	1 160	7 250	740	8 500	**YH-78**	11
12.700	0.5000	17.462	0.6875	6.35	0.2500	4 700	480	2 890	295	5 300	**Y-84**	5.4
		17.462	0.6875	9.52	0.3750	7 650	780	5 350	550	5 300	**Y-86**	7.3
		17.462	0.6875	11.13	—	9 000	915	6 600	675	5 300	**Y-87**	8.2
		17.462	0.6875	12.70	0.5000	10 300	1 050	7 850	800	5 300	**Y-88**	9.1
		17.462	0.6875	15.88	0.6250	12 700	1 300	10 300	1 050	5 300	**Y-810**	11
		17.462	0.6875	19.05	0.7500	15 000	1 530	12 800	1 310	5 300	**Y-812**	14
		19.050	0.7500	15.88	0.6250	14 600	1 480	10 300	1 050	7 500	**YH-810**	16
14.288	0.5625	19.050	0.7500	12.70	0.5000	11 000	1 120	8 750	890	5 000	**Y-98**	9.5
		19.050	0.7500	15.88	0.6250	13 600	1 380	11 500	1 170	5 000	**Y-910**	12
		19.050	0.7500	19.05	0.7500	16 000	1 640	14 200	1 450	5 000	**Y-912**	15
15.875	0.6250	20.638	0.8125	7.92	—	7 050	720	5 050	515	4 500	**Y-105**	6.4
		20.638	0.8125	12.70	0.5000	11 600	1 180	9 600	980	4 500	**Y-108**	11
		20.638	0.8125	19.05	0.7500	17 000	1 730	15 700	1 600	4 500	**Y-1012**	17
		22.225	0.8750	12.70	0.5000	13 300	1 360	9 350	955	6 000	**YH-108**	15
17.462	0.6875	22.225	0.8750	19.05	0.7500	17 900	1 820	17 100	1 740	4 000	**Y-1112**	24
19.050	0.7500	25.400	1.0000	6.35	0.2500	6 250	640	3 650	370	5 300	**Y-124**	8.2
		25.400	1.0000	12.70	0.5000	10 800	1 100	7 350	750	5 300	**Y-128**	17
		25.400	1.0000	15.88	0.6250	18 600	1 890	14 700	1 500	5 300	**Y-1210**	22
		25.400	1.0000	19.05	0.7500	22 100	2 250	18 400	1 880	5 300	**Y-1212**	27
20.638	0.8125	26.988	1.0625	9.52	0.3750	11 300	1 150	7 900	805	5 000	**Y-136**	13
		26.988	1.0625	12.70	0.5000	15 500	1 580	11 900	1 210	5 000	**Y-138**	19
		28.575	1.1250	15.88	0.6250	21 300	2 170	15 500	1 580	6 000	**YH-1310**	28
		28.575	1.1250	19.05	0.7500	25 600	2 610	19 600	2 000	6 000	**YH-1312**	37

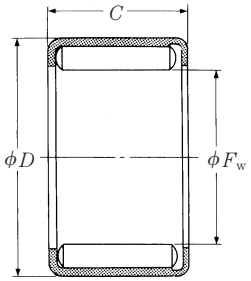
Remarks 1. In this type of bearing, a special grease holds the rollers in place until mounting.
2. Metric series bearings are available. Please consult NSK.

Full Complement Type (Grease-Retained Roller Type)

Y

YH

Inscribed Circle
Diameter 22.225 — 47.625 mm



Boundary Dimensions (mm, inch) F_w		D		$C_{-0.25}^{0}$		Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}	Limiting Speeds (min^{-1}) Oil	Bearing Numbers	Mass (g) approx
22.225	0.8750	28.575	1.1250	12.70	0.5000	15 900	1 620	12 700	1 290	4 500	**Y-148**	20
		28.575	1.1250	19.05	0.7500	23 700	2 410	21 100	2 150	4 500	**Y-1412**	31
		28.575	1.1250	25.40	1.0000	31 000	3 150	29 600	3 000	4 500	**Y-1416**	42
		30.162	1.1875	19.05	0.7500	25 500	2 600	20 500	2 090	5 300	**YH-1412**	38
25.400	1.0000	31.750	1.2500	12.70	0.5000	17 200	1 750	14 300	1 460	4 000	**Y-168**	23
		31.750	1.2500	19.05	0.7500	25 500	2 600	23 900	2 430	4 000	**Y-1612**	35
		31.750	1.2500	25.40	1.0000	33 000	3 400	33 500	3 400	4 000	**Y-1616**	47
		33.338	1.3125	12.70	0.5000	18 800	1 920	13 800	1 400	5 000	**YH-168**	30
		33.338	1.3125	19.05	0.7500	28 600	2 920	23 600	2 410	5 000	**YH-1612**	43
		33.338	1.3125	25.40	1.0000	37 500	3 850	33 500	3 400	5 000	**YH-1616**	58
28.575	1.1250	34.925	1.3750	12.70	0.5000	18 300	1 870	16 000	1 630	3 600	**Y-188**	26
		34.925	1.3750	19.05	0.7500	27 200	2 780	26 600	2 710	3 600	**Y-1812**	38
		34.925	1.3750	25.40	1.0000	35 500	3 600	37 500	3 800	3 600	**Y-1816**	52
30.162	1.1875	38.100	1.5000	15.88	0.6250	26 500	2 700	22 200	2 260	4 300	**Y-1910**	43
31.750	1.2500	38.100	1.5000	15.88	0.6250	24 300	2 480	23 500	2 400	3 200	**Y-2010**	34
		38.100	1.5000	19.05	0.7500	28 900	2 940	29 300	2 990	3 200	**Y-2012**	42
		38.100	1.5000	25.40	1.0000	37 500	3 850	41 000	4 200	3 200	**Y-2016**	57
		38.100	1.5000	28.58	1.1250	41 500	4 250	47 000	4 800	3 200	**Y-2018**	64
		38.100	1.5000	31.75	1.2500	45 500	4 650	53 000	5 400	3 200	**Y-2020**	72
34.925	1.3750	41.275	1.6250	12.70	0.5000	20 000	2 040	19 300	1 960	2 800	**Y-228**	36
		41.275	1.6250	19.05	0.7500	29 700	3 050	32 000	3 250	2 800	**Y-2212**	47
		41.275	1.6250	31.75	1.2500	47 000	4 800	58 000	5 900	2 800	**Y-2220**	76
38.100	1.5000	47.625	1.8750	9.52	0.3750	16 900	1 730	11 600	1 190	4 000	**Y-246**	34
		47.625	1.8750	12.70	0.5000	24 700	2 520	18 900	1 930	4 000	**Y-248**	48
		47.625	1.8750	22.22	0.8750	44 500	4 550	40 500	4 150	4 000	**Y-2414**	89
		47.625	1.8750	25.40	1.0000	51 000	5 200	48 000	4 850	4 000	**Y-2416**	100
		47.625	1.8750	31.75	1.2500	62 500	6 350	62 000	6 350	4 000	**Y-2420**	125
41.275	1.6250	50.800	2.0000	15.88	0.6250	33 000	3 350	27 900	2 840	3 800	**Y-2610**	73
44.450	1.7500	53.975	2.1250	25.40	1.0000	54 500	5 550	55 000	5 600	3 600	**Y-2816**	120
47.625	1.8750	57.150	2.2500	19.05	0.7500	43 000	4 400	41 000	4 200	3 400	**Y-3012**	115

Remarks 1. In this type of bearing, a special grease holds the rollers in place until mounting.
2. Metric series bearings are available. Please consult NSK.

Miniature Bearings with Cages

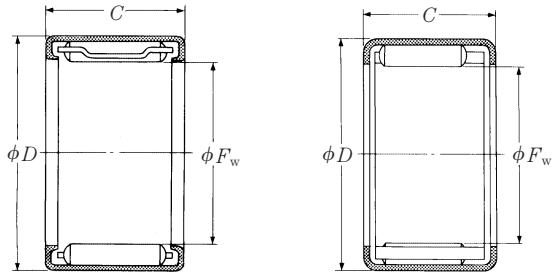
FJ • FJH • FJL (Open Type)

FJP

MFJ • MFJH (Closed Type)

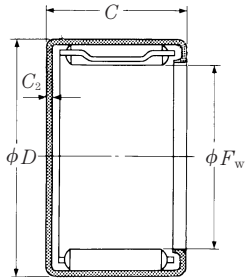
MFJL

Inscribed Circle Diameter 3 — 9 mm



Open Type

FJ, FJH, FJL FJP



Closed Type

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min^{-1})
F_w	D	C $^{0}_{-0.25}$	C_2 max	C_r		P_{max}		Oil
3	6.5	6	—	1 230	125	465	47	100 000
4	8	8	—	1 720	175	670	69	75 000
5	9	8	—	1 550	158	585	60	71 000
	9	9	0.8	1 860	190	745	76	71 000
6	10	9	0.8	2 320	237	985	101	56 000
7	11	9	0.8	2 550	260	1 110	113	48 000
8	12	8	—	2 580	263	1 150	118	40 000
	12	10	0.8	2 840	289	1 270	130	43 000
	14	10	1	4 300	435	1 770	180	45 000
	15	10	1	4 400	450	1 770	181	45 000
	15	15	1	7 100	720	3 250	330	45 000
	15	20	1	10 100	1 020	5 100	520	45 000
9	13	10	0.8	3 300	335	1 600	163	36 000
	15	10	1	4 550	465	1 910	194	40 000
	16	12	1	5 150	525	2 140	219	40 000
	16	16	1	7 800	795	3 650	370	40 000

Bearing Numbers				Mass (g)	
Open		Closed			
FJP, FJ FJH	FJL	MFJ MFJH	MFJL	approx Open	Closed
＊**FJP-36**		—		0.9	—
＊**FJP-48**		—		1.3	—
FJ-58L		—		1.6	—
FJ-59		**MFJ-59**		1.7	1.9
FJ-69		**MFJ-69**		2.2	2.4
FJ-79		**MFJ-79**		2.3	2.7
FJ-88-2		—		2.3	—
FJ-810		**MFJ-810**		2.7	3.2
FJH-810		**MFJH-810**		5.2	5.5
	FJL-810		**MFJL-810L**	6.0	6.6
	FJL-815		**MFJL-815L**	8.9	9.9
	FJL-820		**MFJL-820L**	15	13
FJ-910		**MFJ-910**		3.2	3.6
FJH-910		**MFJH-910**		5.7	6.1
	FJL-912L		**MFJL-912L**	7.7	8.5
	FJL-916		**MFJL-916L**	11	12

＊ These bearings have polyamide cages. The maximum permissible operating temperature for these bearings is 100˚C for continued operation and 120˚C for short periods.

Full Complement Miniature Bearings

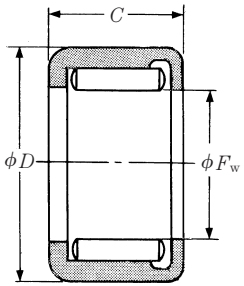
F (Open Type)

MF (Closed Type)

MFY

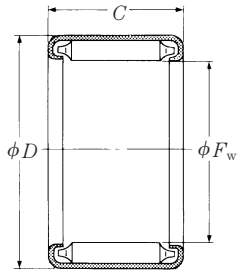
DD (Open Type • Closed Type)

Inscribed Circle Diameter 3.036 — 9 mm



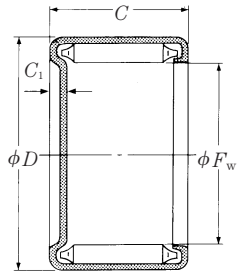
Open Type

DD



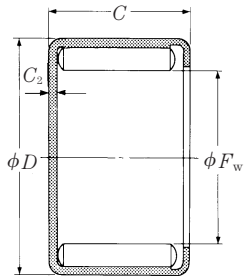
Open Type

F



Closed Type

MF



Closed Type

MFY, DD

Boundary Dimensions (mm)				Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Speeds (min⁻¹)	Bearing Numbers		Mass (g) approx	
F_w	D	C $^{0}_{-0.25}$	C_1, C_2 max	C_r		P_{max}		Oil	Open F, DD	Closed MF, MFY, DD	Open	Closed
3.036	7.035	6.65	—	1 630	167	855	87	12 000	**DD500302**	—	1.4	—
	7.035	9.2	—	2 330	238	1 360	138	12 000	**DD500304**	—	2.0	—
4	8	6	0.8	2 240	229	1 150	117	12 000	—	**DD500401**	—	1.5
	8	8	1.7	2 470	252	1 300	132	12 000	**F-48**	**MF-48**	1.4	1.8
5	9	8	1.7	2 790	285	1 520	155	10 000	**F-58**	**MF-58**	1.8	2.2
	9	7	0.8	3 000	305	1 670	170	10 000	—	**MFY-57**	—	2.1
	9	9	1.7	3 250	335	1 860	190	10 000	**F-59**	**MF-59**	2.0	2.3
6	10	8	1.7	3 150	325	1 780	181	8 500	**F-68**	**MF-68**	2.1	2.7
	10	9	1.7	3 700	380	2 170	221	8 500	**F-69**	**MF-69**	2.3	2.8
7	11	7.5	0.8	4 100	420	2 480	253	7 500	—	**DD500701**	—	2.8
	11	9	1.7	4 100	420	2 480	253	7 500	**F-79**	**MF-79**	2.7	3.2
8	12	8	1.7	3 850	390	2 280	233	7 100	**F-88**	**MF-88**	2.7	3.2
	12	8	0.8	4 750	485	2 990	305	7 100	—	**MFY-88**	—	3.3
	12	10	1.7	5 000	510	3 200	325	7 100	**F-810**	**MF-810**	3.6	4.1
	12	12	1.7	6 200	635	4 200	430	7 100	**F-812**	**MF-812**	3.8	4.7
9	13	9	0.8	5 650	575	3 850	395	6 000	—	**MFY-99**	—	4.1
	13	10	1.7	5 400	550	3 650	370	6 000	**F-910**	**MF-910**	4.1	4.5

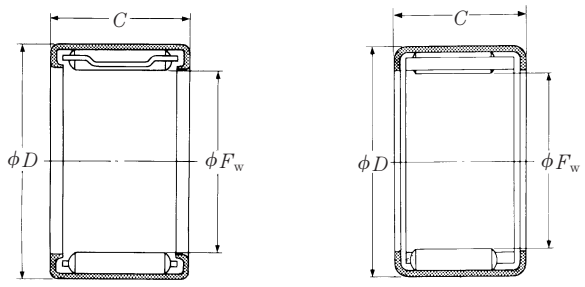
Miniature Bearings with Cages

J • JP (Open Type)

JH

MJ • MJH (Closed Type)

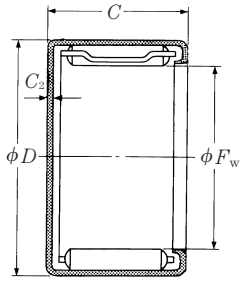
Inscribed Circle
Diameter 3.175 — 9.525 mm



Open Type

J, JH

JP



Closed Type

Boundary Dimensions (mm, inch) F_w		D		C $^{0}_{-0.25}$		C_2 max	Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}	Limiting Speeds (min^{-1}) Oil	Bearing Numbers Open J, JP, JH	Closed MJ, MJH	Mass (g) approx Open	Closed
3.175	0.1250	6.350	0.2500	4.78	—	—	890	91	320	32	75 000	＊**JP-23-FV**	—	0.5	—
3.969	—	7.144	—	4.78	—	—	910	93	320	32	75 000	＊**JP-2** 1/2 **3-FV**	—	0.5	—
4.762	0.1875	8.731	—	4.78	—	—	1 090	111	370	38	75 000	＊**JP-33-FV**	—	0.9	—
		8.731	—	9.52	0.3750	1.0	2 290	234	970	99	75 000	**J-36**	**MJ-361**	2.3	2.7
6.350	0.2500	11.112	0.4375	6.35	0.2500	—	2 040	208	775	79	56 000	＊**JP-44-FV**	—	1.8	—
		11.112	0.4375	7.92	—	1.0	1 940	198	720	74	56 000	**J-45**	**MJ-451**	2.3	2.7
		11.112	0.4375	11.13	—	1.0	3 300	335	1 430	146	56 000	**J-47**	**MJ-471**	3.6	4.1
7.938	0.3125	12.700	0.5000	7.92	—	1.0	2 300	235	900	92	43 000	**J-55**	**MJ-551**	3.0	3.3
		12.700	0.5000	11.13	—	1.0	3 900	400	1 790	183	43 000	**J-57**	**MJ-571**	4.1	4.5
		14.288	0.5625	11.13	—	1.0	4 700	480	1 950	199	45 000	**JH-57**	**MJH-571**	5.9	6.4
9.525	0.3750	14.288	0.5625	7.92	—	1.0	2 620	267	1 080	110	36 000	**J-65**	**MJ-651**	3.2	3.6
		14.288	0.5625	12.70	0.5000	1.0	5 250	535	2 640	270	36 000	**J-68**	**MJ-681**	5.4	5.9

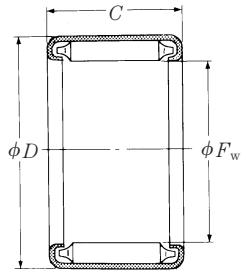
* These bearings have polyamide cages. The maximum permissible operating temperature for these bearings is 100℃ for continued operation and 120℃ for short periods.

Full Complement Miniature Bearings

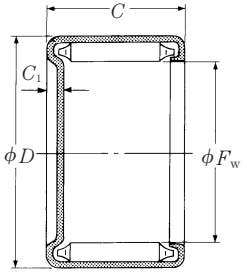
B • BH (Open Type)

M • MH (Closed Type)

Inscribed Circle
Diameter 3.175 — 9.525 mm



Open Type



Closed Type

Boundary Dimensions (mm, inch)							Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}
F_w		D		C $^{0}_{-0.25}$		C_1 max	C_r		P_{max}	
3.175	0.1250	6.350	0.2500	6.35	0.2500	1.3	1 640	167	835	85
3.969	—	7.144	—	6.35	0.2500	1.3	1 930	197	1 000	102
		7.144	—	7.92	—	1.3	2 520	257	1 420	144
4.762	0.1875	8.731	—	6.35	0.2500	1.8	2 100	214	1 040	106
		8.731	—	9.52	0.3750	1.8	3 600	365	2 090	213
6.350	0.2500	11.112	0.4375	6.35	0.2500	2.0	2 490	254	1 200	122
		11.112	0.4375	7.92	—	2.0	3 350	345	1 760	180
		11.112	0.4375	9.52	0.3750	2.0	4 350	440	2 440	249
		11.112	0.4375	11.13	—	2.0	5 250	535	3 150	320
7.938	0.3125	12.700	0.5000	7.92	—	2.0	3 850	395	2 120	216
		12.700	0.5000	9.52	0.3750	2.0	5 000	510	2 930	299
		12.700	0.5000	11.13	—	2.0	6 050	615	3 750	385
		12.700	0.5000	14.27	—	2.0	8 050	820	5 400	555
		14.288	0.5625	11.13	—	2.3	6 650	680	3 700	375
9.525	0.3750	14.288	0.5625	7.92	—	2.0	4 300	440	2 470	252
		14.288	0.5625	9.52	0.3750	2.0	5 550	565	3 400	350
		14.288	0.5625	11.13	—	—	6 750	690	4 400	450
		14.288	0.5625	12.70	0.5000	2.0	7 850	800	5 350	545
		14.288	0.5625	14.27	—	—	8 950	915	6 300	645
		14.288	0.5625	15.88	0.6250	2.0	10 000	1 020	7 300	740
		15.875	0.6250	12.70	0.5000	2.3	8 950	910	5 300	540

Limiting Speeds (min^{-1})	Bearing Numbers				Mass (g)	
	Open		Closed		approx	
Oil	B	BH	M	MH	Open	Closed
12 000	**B-24**		**M-241**		0.9	1.4
10 000	**B-2** 1/2 **4**		**M-2** 1/2 **41**		0.9	1.4
10 000	**B-2** 1/2 **5**		**M-2** 1/2 **51**		1.4	1.8
10 000	**B-34**		**M-341**		1.8	2.3
10 000	**B-36**		**M-361**		2.3	2.7
10 000	**B-44**		**M-441**		2.3	2.7
10 000	**B-45**		**M-451**		2.7	3.2
10 000	**B-46**		**MC-461**		3.4	3.9
10 000	**B-47**		**M-471**		4.1	4.5
8 000	**B-55**		**M-551**		3.6	4.1
8 000	**B-56**		**M-561**		4.5	4.8
8 000	**B-57**		**M-571**		5.0	5.4
8 000	**B-59**		**M-591**		6.4	6.8
11 000		**BH-57**		**MH-571**	6.4	6.8
7 100	**B-65**		**M-651**		4.1	4.5
7 100	**B-66**		**M-661**		5.0	5.4
7 100	**B-67**		—		5.9	—
7 100	**B-68**		**M-681**		6.8	7.3
7 100	**B-69**		—		7.7	—
7 100	**B-610**		**M-6101**		8.6	9.1
10 000		**BH-68**		**MH-681**	9.1	9.5



SOLID NEEDLE ROLLER BEARINGS

METRIC DESIGN

		Pages
RLM · LM, RNA · NA	Inscribed Circle Diameter 8–115 mm	B 96–B109
RNA · NA	Inscribed Circle Diameter 120–490 mm	B110–B113
RNAF · NAF, RNAFW · NAFW	Inscribed Circle Diameter 10–100 mm	B114–B117
RNA···TT, NA···TT	Inscribed Circle Diameter 14–125 mm	B118–B121

INCH DESIGN

HJ, HJ+IR	Inscribed Circle Diameter 15.875–234.950 mm	B122–B127

Solid needle roller bearings are high-accuracy bearings with maximum load capacity within a limited space for various operating conditions.

Made of carefully selected vacuum-degassed bearing steel or carburizing steel, the raceway rings are finished with accurate grinding after heat treatment. The outer rings have strong integrated ribs and contain high-accuracy rollers that have the proper crowning finish. Lightweight and extremely strong, the cage guides the rollers accurately and smoothly.

DESIGN AND TYPES

Solid needle roller bearings come in various types ranging from the light to heavy load use of the metric bearings to heavy load use of inch bearings. Each type is available in series with and without an inner ring, and both series are shown in the same bearing table. Types of solid needle roller bearings are shown in **Table 1**, while the composition of bearing numbers is shown in **Table 2**.

Among NSK solid needle roller bearings, the RNA, NA49, 59, 69 and 48 series follow the international dimension system that conforms to ISO standards.

HJ, HJ + IR series of inch bearings are widely used globally and conform to the MIL standards of the U.S.A.

RLM and LM types are bearings whose section height is close to that of drawn cup needle bearings.

RNAF and NAF are different from other solid bearings in that they have a separable inner ring, outer ring, and cage and rollers, which can all be combined to allow the optional radial internal clearance to be chosen. For most operating conditions, they can be easily mounted.

Bearings with seals are also manufactured; please consult NSK.

Typical structure and features of solid needle roller bearings are shown in **Fig. 1**.

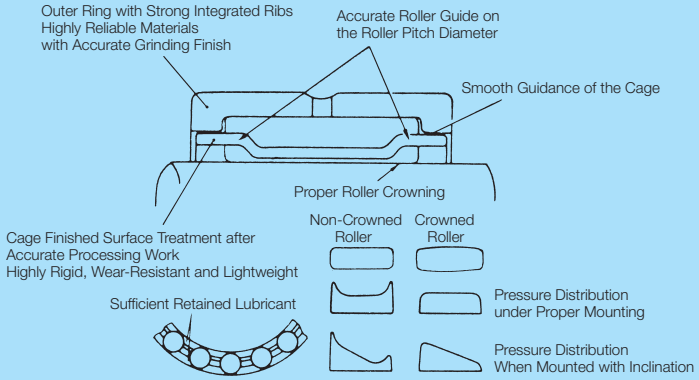


Fig. 1 Features of Solid Needle Roller Bearings

Table 1 Bearing Types

Bearing Code			Shaft Diameter (mm)	Description
Series	Without Inner Ring	With Inner Ring		
Metric	RLM		8 to 110	Single-row, section height less than or equal to dimensional series 49, for light loads
		LM	5 to 90	
	RNA49		8 to 490	Single-row, dimensional series 49, for normal loads
		NA49	8 to 440	
	RNA59		20 to 160	Single-row, dimensional series 59, same section height as series 49 with wider width, for heavy loads
		NA59	15 to 140	
	RNA69		16 to 110	Single-row, dimensional series 69, same section height as series 49, wider than series 59, for heavy loads
		NA69	12 to 95	
	RNA48		120 to 415	Single-row, dimensional series 48, smaller section height than series 49, for light to normal loads
		NA48	110 to 380	
	RNAF, RNAFW		10 to 100	Single or double-row, separate type, for normal to heavy loads
		NAF, FAFW	10 to 90	
	RNA···TT	NA ... TT	10 to 120	With seal
Inch	HJ		15.875 to 234.950	Single-row, section height slightly greater, with wider width series, for heavy loads
		HJ + IR	9.525 to 203.200	

Table 2 Bearing Numbers for Solid Needle Roller Bearings

Type Code		Composition of Bearing Number	Examples
Without Inner Ring	With Inner Ring		
RLM		Type code, inscribed circle diameter, and width or code, inscribed circle diameter, outside diameter, and width (expressed in mm)	**RLM2520**, **RLM304020**
	LM		**LM2520**, **LM304020**
RNA		Type code, dimension series, and bore number (48, 49, 59, 69) (00 ~ 88)	**RNA4905**, **RNA6908**, **RNA4830**
	NA		**NA4905**, **NA6908**, **NA4830**
RNAF		Type code, inscribed circle diameter, outside diameter, and width (expressed in mm)	**RNAF253517**
	NAF	Type code, bore diameter, outside diameter, and width (expressed in mm)	**NAF203517**
HJ		Type code, hyphen, inscribed circle diameter, outside diameter, and width (expressed in integers in units of 1/16 inch)	**HJ-243316**
	HJ + IR	Following HJ bearing number, + inner ring code, hyphen, bore diameter, inner ring raceway diameter, and width (expressed in integers in units of 1/16 inch)	**HJ-243316+IR-202416**

Remarks For full complement RLM, LM, RNA and NA, the letter "V" is followed by the bearing number.

ACCURACY

All NSK solid needle roller bearings, excluding inch bearings HJ and HJ + IR, conform to ISO, and dimensional accuracy and rotating accuracy are in accordance with ISO (refer to pages A26 and A27). When highly accurate bearings are especially required, bearings in Class 6 or higher are also available.

Tolerance values for inch bearings HJ and HJ + IR are shown in **Table 3**.

Tolerances for inscribed Circle Diameter

NSK metric needle roller bearings are primarily manufactured within the inscribed circle diameter tolerance class F6 (refer to **Table 4** on page B92). RLM and LM types that have a small section height, however, are whithin F7. Tolerances for the inscribed circle diameter of inch needle bearings are shown in **Table 5** on page B93.

Table 3 Tolerances for Inch Bearings HJ, HJ + IR

Units: μm

Nominal Bore Diameter, d, or Nominal Outside Diameter, D (mm)		Single Plane Mean Bore Diameter Deviation, Δ_{dmp}		Single Plane Mean Outside Diameter Deviation, Δ_{Dmp}		Inner Ring Width Deviation, Δ_{Bs}		Outer Ring Width Deviation, Δ_{Cs}		Radial Runout of Assembled Inner Ring, K_{ia}	Radial Runout of Assembled Outer Ring, K_{ea}
over	incl	high	low	high	low	high	low	high	low	max	max
7.938 (5/16)	19.050 (3/4)	0	−10	—	—	+250	+120	—	—	10	—
19.050 (3/4)	30.162 (1 3/16)	0	−13	0	−13	+250	+120	0	−130	13	15
30.162 (1 3/16)	50.800 (2)	0	−13	0	−13	+250	+120	0	−130	15	20
50.800 (2)	82.550 (3 1/4)	0	−15	0	−15	+250	+120	0	−130	20	25
82.550 (3 1/4)	107.950 (4 1/4)	0	−20	0	−20	+250	+120	0	−130	25	35
107.950 (4 1/4)	120.650 (4 3/4)	0	−20	0	−20	+380	+250	0	−130	25	35
120.650 (4 3/4)	177.800 (7)	0	−25	0	−25	+380	+250	0	−130	30	45
177.800 (7)	184.150 (7 1/4)	0	−30	0	−25	+380	+250	0	−130	30	45
184.150 (7 1/4)	203.200 (8)	0	−30	0	−30	+380	+250	0	−130	40	50
203.200 (8)	260.350 (10 1/4)	—	—	0	−30	—	—	0	−130	—	50
260.350 (10 1/4)	317.500 (12 1/2)	—	—	0	−36	—	—	0	−130	—	60

Note Δ_{dmp}, Δ_{Bs} and K_{ia} are found from the dimension category of d, and Δ_{Dmp}, Δ_{Cs} and K_{ea} are from the dimension category of D.

Table 4 Inscribed Circle Diameter for Metric Solid Needle Roller Bearings

Nominal Inscribed Circle Diameter, F_w (mm)		Deviation (F6) of Minimum Diameter, $F_{w\,min}$, of Roller Inscribed Circle Diameter (1) (μm)	
over	incl	high	low
6	10	+ 22	+13
10	18	+ 27	+16
18	30	+ 33	+20
30	50	+ 41	+25
50	80	+ 49	+30
80	120	+ 58	+36
120	180	+ 68	+43
180	250	+ 79	+50
250	315	+ 88	+56
315	400	+ 98	+62
400	500	+108	+68

Note (1) When using a cylinder instead of an inner ring, $F_{w\,min}$ is the diameter of the cylinder at which the internal clearance is zero in at least one radial direction. ($F_{w\,min}$ is the minimum diameter of each inscribed circle diameter where deviation is suspected.)

Table 5 Tolerance of Inscribed Circle Diameter for Inch Solid Needle Roller Bearings

Nominal Inscribed Circle Diameter, F_w (mm)		Deviation (F6) of Minimum Diameter, $F_{w\,min}$, of Roller Inscribed Circle Diameter (1) (μm)	
over	incl	high	low
—	15.875 (5/8)	+43	+20
15.875 (5/8)	28.575 (1 1/8)	+46	+23
28.575 (1 1/8)	41.275 (1 5/8)	+48	+25
41.275 (1 5/8)	47.625 (1 7/8)	+51	+25
47.625 (1 7/8)	69.850 (2 3/4)	+53	+28
69.850 (2 3/4)	76.200 (3)	+58	+28
76.200 (3)	101.600 (4)	+61	+30
101.600 (4)	114.300 (4 1/2)	+66	+30
114.300 (4 1/2)	152.400 (6)	+69	+33
152.400 (6)	165.100 (6 1/2)	+74	+33
165.100 (6 1/2)	196.850 (7 3/4)	+76	+36
196.850 (7 3/4)	234.950 (9 1/4)	+81	+36

Note (1) When using a cylinder instead of an inner ring, $F_{w\,min}$ is the diameter of the cylinder at which the internal clearance is zero in at least one radial direction. ($F_{w\,min}$ is the minimum diameter of each inscribed circle diameter where deviation is suspected.)

APPLICATION

Specifications of Shaft and Housing

The shaft and housing bore should be finished with recommended dimensional tolerance. Accuracy, surface roughness, and hardness should satisfy the conditions shown in **Table 6**.

Shaft Inclination

Shaft inclination due to deflection by external force and mounting error of the bearing should be the same as or less than the values found in **Table 7**.

Table 6 Accuracy and Hardness of Shaft and Housing

Category	Shaft		Housing Bore
	Raceway Surface	Fitting Surface	
Out-of-Roundness Tolerance	$\frac{IT3}{2}$	$\frac{IT3}{2}$ to $\frac{IT4}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Cylindrical Tolerance	$\frac{IT3}{2}$	$\frac{IT3}{2}$ to $\frac{IT4}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Roughness, Ra	0.4	0.8	1.6
Hardness	HRC58 to 64 Appropriate depth of hardening layer required	—	—

Note (1) Please refer to Section **5.2** on page A34 and Section **5.3** on page A35 for further details.

Table 7 Shaft Inclination

Bearing Width (mm)		Permissible Inclination (mm/mm)
over	incl	With cage
—	25	0.0015
25	50	0.0010
50	—	0.0005

Fitting

Depending on the application, the outer ring of NSK solid needle roller bearings is used with either a transitional fit or a tight fit.

Recommended fittings for solid needle roller bearings are shown in **Table 8**.

Please refer to **Tables 5.2** and **5.3** on page A33 to fully consider operating conditions and choose the appropriate fittings.

For needle bearings without an inner ring, various radial inner clearances are achieved by selecting the proper shaft tolerance from **Table 4.6** on page A25.

Table 8 Recommended Fittings for Solid Needle Roller Bearings

Units: mm

Code		Fitting Tolerance			
		Shaft or Inner Ring Rotation		Outer Ring Rotation	
		Shaft	Housing Bore	Shaft	Housing Bore
RLM, RNA RNAF HJ	Without Inner Ring	h5 ($F_w \leqq 80$) g6 ($180 \geqq F_w > 80$) f6 ($F_w > 180$)	H7	g5 ($F_w \leqq 80$) f6 ($180 \geqq F_w > 80$) e6 ($F_w > 180$)	N7
LM, NA NAF HJ+IR	With Inner Ring	k5 ($d \leqq 50$) m5 ($d > 50$)	H7	g6	N7

Remarks For metric needle bearings with an inner ring, if the inner and outer rings are fitted tighter than class k and K, respectively, a bearing with an internal clearance greater than CN clearance should be selected.

Mounting

Solid needle roller bearings generally require the shaft and housing shoulder to fix the inner and outer rings and prevent them from moving in the axial direction during operation (**Fig. 2**).

The unmarked surface of both the inner and outer rings should be placed onto the shaft or housing shoulder when mounting. The diameter of the shaft shoulder should be smaller than the outside diameter of the inner ring. Because the RNAF and NAF types do not have ribs, abutment and fillet dimensions should be carefully observed when designing the bearing surroundings. Examples of mounting, abutment and fillet dimensions are shown in **Fig. 3** and **Table 9**.

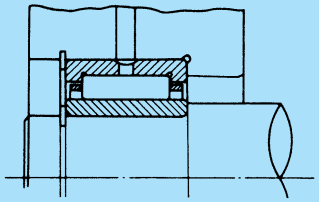


Fig. 2 Example of Mounting

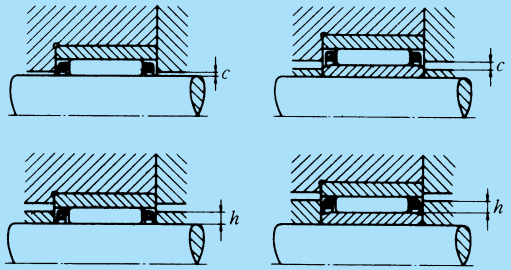


Fig. 3 Example of Mounting for RNAF and NAF

Table 9 Abutment and Fillet Dimensions for RNAF and NAF

Units: mm

Nominal Inscribed Circle Diameter, F_w		Category of D/F_w	c	h
over	incl			
8	12	$D/F_w < 2.0$ $D/F_w \geqq 2.0$	0.15 0.3	1.35 2.8
12	22	$D/F_w < 1.59$ $D/F_w \geqq 1.59$	0.2 0.3	1.8 2.8
22	30	$D/F_w < 1.48$ $D/F_w \geqq 1.48$	0.3 0.4	2.2 3.2
30	55	$D/F_w < 1.34$ $D/F_w \geqq 1.34$	0.3 0.5	2.2 3.6
55	100	—	0.5	3.6